



## MASTER THESIS

# From Retrieval Alignment to Realised Utility

Evaluating Experience Retrieval for an LLM Agent in  
TextWorldExpress CookingWorld

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# Abstract

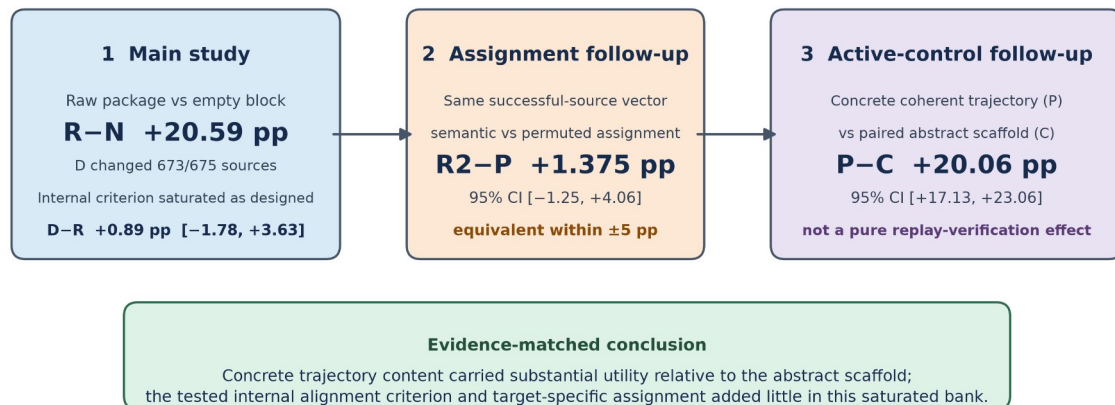
Experience retrieval can improve language-agent performance without updating model parameters, but better offline matching need not imply better interactive behaviour. This thesis asks whether interventions that saturate an internal correspondence criterion, or that restore target-specific assignment, also change downstream utility, and whether utility depends on concrete trajectory content. An experience is a replay-verified environment-generated gold trajectory that ends in official success; it is not a trajectory previously produced by the acting model. Experiments used a frozen bank of such trajectories in TextWorldExpress CookingWorld. The retained main study compared No Memory (N), Raw Semantic retrieval (R), and De-lexicalised Semantic retrieval (D) across 675 targets and 4,044 accepted episodes under one acting model and prompt configuration.

The complete Raw Semantic package improved terminal success by 20.59 percentage points relative to No Memory. De-lexicalised retrieval changed the selected source on 673 of 675 targets and saturated the internal exact-anonymised-correspondence criterion (6.52% under R versus 100% under D), yet differed from Raw Semantic retrieval by only +0.89 percentage points in terminal success (95% CI  $[-1.78, +3.63]$ ). Because the criterion shares the anonymised representation used by the De-lexicalised retriever, it is treated as an internal manipulation check rather than independent ground truth of retrieval quality.

Two sequential follow-ups were specified and frozen before their own outcomes, after the preceding results were known. A first follow-up compared target-specific Raw Semantic assignment with a distribution-matched permutation of the same selected-source vector (P); the estimated assignment increment was +1.375 points (95% CI  $[-1.25, +4.06]$ ), equivalent within the prospectively specified  $\pm 5$ -point margin. A later active-control study compared the concrete permuted trajectory (P) with a paired abstract, format-, length-, and phase-matched scaffold (C); P exceeded C by 20.06 points (95% CI  $[+17.13, +23.06]$ ). The results support a bounded alignment–utility gap: saturating the internal correspondence criterion or restoring target-specific assignment changed terminal success little, whereas the concrete coherent trajectory substantially outperformed the implemented scaffold. This evidence is local to the tested bank and control; P–C does not isolate replay verification or imply that arbitrary trajectories are interchangeable.

## Why did retrieved experience help?

Three sequential studies, each frozen before its own outcomes



Graphical abstract. Three sequential studies narrowed the source of realised utility. The Raw Semantic package improved success over an empty block by +20.59 percentage points. De-lexicalisation changed 673 of 675 selected sources and saturated its internal criterion as designed, but added only +0.89 points over Raw Semantic retrieval. A separately frozen assignment control found a +1.375-point semantic-assignment increment that was equivalent within ±5 points; a later active-control follow-up found a +20.06-point advantage for the concrete trajectory over the paired abstract scaffold. The follow-ups were frozen after the preceding results and do not identify a pure replay-verification effect.

## Dansk resumé

Genfinding af tidligere erfaringer kan forbedre sprogagenters præstation, men bedre offline-matchning medfører ikke nødvendigvis bedre interaktiv adfærd. Denne afhandling undersøger dette spørgsmål i TextWorldExpress CookingWorld med No Memory (N), Raw Semantic retrieval (R) og De-lexicalised Semantic retrieval (D). Hovedanalysen omfatter 675 targets og 4.044 accepterede episoder med én agentmodel og promptkonfiguration.

Den komplette Raw Semantic-hukommelsespakke øgede terminal succes med 20,59 procentpoint i forhold til en tom hukommelsesblok. De-lexicalised retrieval ændrede den valgte kilde for 673 af 675 targets og mættede som tilsigtet sit interne tilpasningskriterium, men terminal succes afveg kun med +0,89 procentpoint fra Raw Semantic retrieval (95 % KI  $[-1,78; +3,63]$ ). Manipulationen blev dermed aktiveret, men det interne kriterium forklarede ikke den store fordel ved hukommelsespakken.

Hovedresultatet motiverede to sekventielle opfølgningsforsøg, som hver blev specificeret og fastfrosset før egne resultater. I det første tilføjede targetspecifik semantisk tildeling 1,375 procentpoint (95 % KI  $[-1,25; +4,06]$ ) og opfyldte ækvivalenskriteriet på  $\pm 5$  procentpoint. I det andet overgik den konkrete permuterede bane (P) et parret abstrakt scaffold (C), som var matchet med hensyn til format, længde og faser, med 20,06 procentpoint (95 % KI  $[+17,13; +23,06]$ ). Evidensen peger derfor på konkret, sammenhængende baneindhold som en vigtig kilde til nytte i denne opsætning, mens targetspecifik tildeling bidrog lidt. P–C identificerer dog ikke en ren effekt af replay-verifikation og viser ikke, at vilkårlige baner er udskiftelige.

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# 1. Introduction

## 1.1 From retrieval alignment to realised utility

This thesis studies LLM agents in text-based interactive task environments, specifically TextWorldExpress CookingWorld (Jansen & Côté, 2023). A CookingWorld task places ingredients and kitchen objects across a small set of locations and gives the agent a cookbook specifying how each ingredient must be prepared before the meal can be consumed. At each step, the agent receives a natural-language observation and a list of currently admissible actions, selects one action, and receives the resulting observation. An episode begins from a fresh task state and ends when the agent completes the recipe and consumes the meal, reaches another terminal state, or exhausts the action limit. A trajectory records the task specification, successive observations, chosen actions, and outcome of one episode. Trajectory-based memory systems store such records from earlier tasks, retrieve a candidate for a new task, and place it in the model context as an in-context execution example (Zhao et al., 2024; Xiong et al., 2026). Because each action changes the next state and available actions, an early error can make the goal harder or impossible to reach.

A trajectory may be successful in its source task yet provide poor guidance in a new target task. Case-based reasoning separates retrieving a prior case from reusing it to address a new problem; its classic cycle further distinguishes revision and retention (Aamodt & Plaza, 1994). The present system implements only a limited analogue of the first two stages. A retrieval policy selects one prior successful trajectory, after which the LLM may draw on that trajectory while choosing actions in the target episode; there is no explicit adaptation module. Offline diagnostics therefore evaluate the retrieval decision: whether two policies select different sources and whether the selected source meets the predefined correspondence criterion. Interactive evaluation instead measures realised utility: whether exposing the agent to the selected trajectory changes terminal success. Neither level of evidence answers the other. A corresponding source may provide no behavioural benefit, while terminal success does not reveal whether, or how, the agent used the trajectory.

Prior exemplar-selection work addresses the first question. EPR learns which demonstrations are useful to a downstream language model rather than relying only on off-the-shelf similarity (Rubin et al., 2022), while STARE uses structural alignment to guide exemplar selection in static reasoning tasks (Li et al., 2025). Research on agent memory addresses the second question. Stored experience can improve subsequent performance (Zhao et al., 2024), while Experience-Following shows that retrieved records can also propagate misaligned execution patterns (Xiong et al., 2026). What remains

unclear is whether an offline advantage in experience selection translates into higher terminal success when the selected trajectory must support a complete interactive episode.

This thesis studies why retrieved experience changes realised utility and which parts of the retrieval package account for that change. Retrieval alignment is defined internally as exact equality of anonymised ordered ingredient–operation chain multisets under the frozen CookingWorld representation. The De-lexicalised retriever uses the same ingredient-anonymised abstraction, so saturation of this criterion is constructible by design. It verifies that D behaved as intended; it is not independent evidence that D selected objectively better or more useful trajectories. Realised utility is instead the target-level change in official terminal success when a retrieval policy is deployed.

The investigation proceeded sequentially. The retained main study compared No Memory (N), Raw Semantic retrieval (R), and De-lexicalised Semantic retrieval (D), testing memory access and a representation change. After those outcomes were known, a separately frozen assignment experiment tested whether target-specific semantic matching mattered. Its result then motivated a separately frozen active-control follow-up comparing concrete trajectory content with an abstract worked-protocol scaffold. Chapter 1 unifies these studies under the question of why retrieved experience helps, but does not recast them as a single decomposition planned in advance.

Table 1.1 separates the quantities that must remain distinct throughout the thesis. In particular, a high alignment score is an offline property of a selected source, whereas realised utility is an outcome difference produced when a retrieval policy is deployed.

**Table 1.1. Core constructs, operational measures, and claim boundaries.**

| <b>Construct</b>     | <b>Operational measure in this thesis</b>                      | <b>What the measure supports</b>                                   | <b>What it does not establish</b>                         |
|----------------------|----------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| Retrieval activation | Final-source agreement and source-change rate between policies | Whether two retrieval conditions actually create different prompts | That the changed source is better or is used by the agent |



| Construct                         | Operational measure in this thesis                                            | What the measure supports                                                       | What it does not establish                                                 |
|-----------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Study-defined retrieval alignment | Exact equality under the frozen anonymised chain representation shared with D | Whether D saturates its intended internal selection criterion                   | Independent correspondence validation, general usefulness, or agent uptake |
| Realised utility                  | Target-level difference in official terminal success                          | Average downstream effect of deploying one retrieval policy rather than another | Why an individual episode succeeded or failed                              |
| Mechanism evidence                | Tie diagnostics, process traces, failure taxonomy, and overlap strata         | Descriptive localisation of where differences may arise                         | Causal mediation or memory uptake by an individual episode                 |

Note. A zero on the exact-correspondence indicator means only that the selected source does not reproduce the complete anonymised target recipe. It does not mean that source and target have no partial structural overlap or that the source has zero downstream utility.

## 1.2 Why CookingWorld

Testing this question in an open-ended agent environment creates an identification problem. A retrieved trajectory can differ from the target in topic, length, state distribution, language, and procedure, while its true applicability is rarely available as an objective label. CookingWorld provides an unusually controlled setting in which retrieval alignment and realised utility can be measured separately. Its cookbook specifies the procedure before execution, and the environment supplies an official terminal-success signal after execution (Jansen & Côté, 2023).

The cookbook contains two distinguishable sources of retrieval evidence. Ingredient names provide entity-level lexical cues, while transformations such as chopping, frying, slicing, and grilling form ordered ingredient-specific chains. Suppose a target requires carrot: chop → fry and potato: slice → grill. A source requiring carrot: chop → grill and potato: slice → fry shares the ingredient names and operation vocabulary but not their procedural allocation. Conversely, a source using different

ingredients with the chains chop → fry and slice → grill changes the surface vocabulary while preserving the recipe structure. These relations are extracted by a deterministic parser rather than inferred by an annotator or another language model.

Every experimental condition receives the complete current cookbook. This rules out improvement from revealing otherwise missing recipe information. A retrieved trajectory may still provide useful execution scaffolding, while a source-specific pattern may help or interfere with later actions. Cookbook visibility therefore makes the experiment a conservative test of retrieval-policy utility under a fully specified target and prevents hidden target information from explaining any observed benefit.

Other TextWorldExpress environments emphasise placement, navigation, arithmetic, sorting, or direct instruction following and would require a different definition of structure. CookingWorld is selected because it combines explicit recipes, reproducible task generation with disjoint folds, replayable successful trajectories, and an objective terminal outcome. The simple two-ingredient, three-location configuration further limits navigation and search variance that could mask a retrieval effect. CookingWorld is therefore used to study structure-sensitive experience retrieval, not as a proxy for general agent intelligence.

### **1.3 Research questions and experimental logic**

The retained main-study analysis uses three target-blocked retrieval configurations. No Memory (N) supplies the target without a past trajectory. Raw Semantic retrieval (R) selects a trajectory from the original cookbook text. De-lexicalised Semantic retrieval (D) applies the same dense encoder after ingredient names are anonymised. Once a source is selected, both memory conditions present it through the same raw-trajectory renderer.

Two prespecified contrasts remain central to the revised analysis. R–N contrasts the complete Raw Semantic memory package with an empty memory block; it does not isolate trajectory-specific experience from generic support supplied by any non-empty worked interaction example. D–R measures the change associated with ingredient anonymisation and is the direct comparison between a retrieval representation that saturates the study-defined alignment criterion and the Raw Semantic baseline. Both were secondary contrasts in the frozen protocol and are reported with their original Holm-adjusted inference; neither is relabelled as confirmatory.

A positive R–N shows that adding the complete Raw Semantic package improved terminal success relative to the empty block, without identifying which bundled component produced the gain. D–R

asks a narrower question: after de-lexicalisation changed almost every selected source and saturated exact anonymised correspondence, did terminal success change as well? The answer is interpreted from its effect estimate and confidence interval rather than from non-significance alone.

RQ1 asks whether De-lexicalised Semantic retrieval changes terminal success relative to Raw Semantic retrieval; its effect on the study-defined offline alignment criterion is reported only as a manipulation check. RQ2 asks whether the complete Raw Semantic memory package improves terminal success relative to an empty memory block. RQ3 asks whether target-specific Raw Semantic assignment adds utility relative to a distribution-matched permutation of the same selected-source vector. RQ4 asks whether a concrete trajectory adds utility beyond the paired abstract scaffold. The retained main-study comparisons were prospectively specified secondary analyses; the thesis preserves that evidential status.

Terminal success is the primary outcome. The offline diagnostics specified in Section 4.6 document whether and how the intervention changed experience selection. They do not diagnose whether an individual failure was caused by unsuccessful application of the retrieved trajectory.

## **1.4 Contribution and scope**

The thesis makes three contributions:

Empirically, the thesis establishes a sequential evidence chain. The complete Raw Semantic package improved terminal success by 20.59 points over an empty block. Most of that benefit remained when the successful-source vector was permuted; the estimated target-specific assignment difference was +1.375 points (95% CI  $[-1.25, +4.06]$ ) and met the frozen  $\pm 5$ -point equivalence criterion. A later active-control follow-up found that the concrete permuted trajectory exceeded its paired abstract scaffold by 20.06 points. These results locate substantial utility in concrete coherent trajectory content relative to the implemented scaffold, while placing a small practical bound on the tested target-specific assignment increment.

Methodologically, the thesis shows how sequential, prospectively frozen follow-ups can narrow a package effect without pretending that the entire decomposition was planned before the first outcomes. The main study, assignment control, and active-control follow-up retain their separate temporal status, target populations, estimands, and claim boundaries.

Diagnostically, the De-lexicalised retriever changed 673 of 675 selected sources and saturated its internal criterion as intended, yet its terminal-success increment over Raw Semantic retrieval was

+0.89 points (95% CI [-1.78, +3.63]). The informative result is not the magnitude of the internal alignment score, but the failure of that activated manipulation to explain the observed package utility. The bank contained 3,121 trajectories but only 136 distinct anonymised recipe structures, termed recipe signatures, so exact matches were generally available. Yet R met the exact anonymised-correspondence criterion on only 44 of 675 targets. The conclusions therefore concern abundant candidate coverage, not a Raw Semantic retriever already selecting matched sources; they neither identify replay verification as the cause of P-C nor generalise beyond the tested model, prompt, Top-1 policy, and CookingWorld setting.

Chapter 2 reviews experiential memory, exemplar selection, and the distinction between surface and structural similarity. Chapter 3 develops the retained N/R/D design and its claim boundaries. Chapter 4 specifies the environment, retrievers, agent interface, and analysis. Chapter 5 reports the results, Chapter 6 discusses their implications and limitations, Chapter 7 addresses ethical and broader considerations, and Chapter 8 concludes the thesis.

## 2. Background and Related Work

### 2.1 Experience retrieval in language agents

External memory allows an agent to reuse earlier interaction without retraining its policy. The stored material may be a full episode, a shorter action sequence, a verbal reflection (Shinn et al., 2023), or an executable skill (Wang et al., 2024). This thesis works with a gold-demonstration bank: replay-verified environment-generated trajectories linked to particular source tasks, rather than episodes previously produced by the acting LLM. The setup resembles case-based reasoning, where solving a new problem begins by finding and adapting a previous case (Aamodt & Plaza, 1994).

ExpeL collects task experience and extracted natural-language insights for later recall (Zhao et al., 2024). ExpRAG builds an index of earlier trajectories and studies practical choices such as query construction, index content, and retrieval depth (Ferraz et al., 2026). These systems show that past interaction can be reused, but they also change several parts of the memory pipeline at once.

Text games provide a controlled setting for studying such systems because they require language understanding and multi-step problem solving while retaining reproducible environment dynamics (Jansen & Côté, 2023). Results from other text-game tasks do not provide a directly transferable success rate for the simplified CookingWorld configuration used here. Its expected success range and run-to-run variation were therefore estimated on excluded pilot targets rather than borrowed from a different task or policy stack.

Experience-Following isolates a risk that is especially relevant to the present work. Similar current and stored inputs often lead to similar executions; the authors identify error propagation and misaligned replay as consequences of this tendency (Xiong et al., 2026). A trajectory can be successful in its own task yet be misleading elsewhere. Source validation and source applicability are therefore different requirements.

The experiment holds memory construction fixed. Its bank contains replay-verified successes and remains read-only during evaluation. It does not test online addition, deletion, consolidation, or learned memory policies. This removes features that matter in deployed systems, but ensures that R and D choose from exactly the same candidates.

### 2.2 Dense retrieval and exemplar selection

This thesis treats a retrieved trajectory as an in-context exemplar for the acting policy. The analogy is useful but not exact. Demonstrations in much of the in-context learning literature support a single

model response; an interaction trajectory is longer, contains environment feedback, has internal temporal order, and must support decisions across many subsequent steps. Whether findings about single-turn demonstration selection carry over to this temporally structured object is part of the question tested here.

The choice of an in-context example can materially affect a language model even when the model and task remain unchanged. KATE retrieves demonstrations by semantic similarity and reports consistent gains over random selection across language-understanding and generation tasks (Liu et al., 2022). EPR goes further by using language-model scores to label useful demonstrations and training a dense retriever from those preferences (Rubin et al., 2022). These approaches differ in supervision, but both treat exemplar selection as a retrieval problem rather than a fixed prompt-design choice. The wider in-context learning literature includes random, similarity-based, diversity-aware, and learned selection strategies (Dong et al., 2024).

More recent work extends exemplar selection beyond general semantic similarity or learned downstream preference. SGA-MCTS uses entity abstraction in experience retrieval, while STARE uses structural alignment to guide exemplar selection (Xie et al., 2026; Li et al., 2025). These non-interactive or differently structured settings motivate testing whether a representation that suppresses entity identity changes which trajectory is selected and whether that change survives interactive execution.

Dense retrieval is therefore a substantive baseline rather than a straw man. It is efficient, reproducible, and stronger than lexical overlap alone: target and candidate descriptions are encoded as vectors and ranked by cosine similarity. A general-purpose encoder may already capture operation order and role assignment. Comparing raw and de-lexicalised inputs tests whether suppressing ingredient identity adds useful selection information beyond this baseline.

The experiment does not train a task-specific retriever. The same frozen encoder is used for R and D, leaving query representation as the planned difference. This gives up the possible advantage of a learned retriever such as EPR or STARE, but avoids introducing training data and optimisation choices that would obscure the comparison.

## **2.3 Surface similarity and structural correspondence**

The distinction between surface and structural similarity predates contemporary retrieval systems. Structure-mapping theory describes analogy in terms of correspondences between relations rather than matches between object attributes (Gentner, 1983). Later work treats analogy and literal

similarity as points within a broader process of structural alignment: literal similarity may preserve both attributes and relations, whereas analogy can preserve relational organisation despite different entities (Gentner & Markman, 1997).

Experimental research on analogical transfer also shows why retrieval cannot be reduced to structural fit alone. Gick and Holyoak (1983) found that abstracting a common schema from earlier examples could support later problem solving. Yet possessing a useful source did not ensure spontaneous transfer: participants were substantially more likely to use it when its relevance to the target was made explicit. Access in the present experiment is guaranteed by construction because the selected trajectory appears in the prompt. The agent therefore resembles a participant in the hint condition rather than one required to retrieve the precedent unaided. What remains open is whether providing one source rather than another changes realised utility. Terminal success does not by itself establish that the agent applied the precedent.

Holyoak and Koh (1987) showed that both structural relations and salient surface features influence analogical transfer. Surface overlap is not merely noise; it can provide an effective access cue to a useful source. This produces competing predictions for the D–R comparison. Ingredient anonymisation may help by preventing entity overlap from dominating the ranking, but it may also hurt by removing lexical evidence that usefully narrows the search. Whether ingredient identity is informative in generated CookingWorld tasks depends partly on the empirical association between ingredients and operation chains in the frozen bank. That association is reported as an outcome-independent material diagnostic. The direction of D–R is not assumed in advance.

CookingWorld provides a limited computational version of this distinction. Ingredient identity is treated as a surface cue, while the assignment and order of transformations within ingredient-specific chains define the study's procedural representation. For the target carrot: chop → fry, a source mentioning carrot but requiring slice → grill shares an entity without sharing the procedure. A source using potato with chop → fry reverses that pattern. Raw text contains both kinds of evidence, whereas de-lexicalised text suppresses ingredient identity.

“Surface cue” and “procedural structure” name retrieval representations in this thesis; they are not psychological measurements. The experiment does not test whether the agent spontaneously notices an analogy, induces a schema, maps higher-order relations, or exhibits Gentner’s systematicity principle. It asks the narrower engineering question of whether a retriever that represents a small procedural relation selects more useful prior trajectories. A positive result would support structure-

sensitive experience selection in this environment, not human-like analogical reasoning by the acting model.

## 2.4 Evaluating retrieval alignment and realised utility

As defined in Section 1.1, retrieval alignment is operationalised through the frozen set of offline measures specified in Section 4.6. Realised utility is estimated from differences in terminal success on the same targets. The first set of measurements describes what a retrieval policy selects under the predefined recipe representation; the second estimates what changes when that policy is deployed.

This distinction matters because an offline match and an interactive outcome answer different questions. De-lexicalisation may improve alignment without changing behaviour if the agent already knows how to solve the task, ignores the retrieved trajectory, or cannot adapt it during execution. It may also remove lexical cues that were useful even when they were classified as surface information.

Neither a retrieval diagnostic nor an individual episode is sufficient to estimate utility. The de-lexicalised method partly defines alignment through its own representation, and two hosted-model executions can diverge even when their prompts are identical. The study therefore uses outcome-blind retrieval diagnostics, target-level blocked outcome contrasts, and repeated executions while keeping alignment, utility, and mechanism evidence conceptually separate.

The literature reviewed in this chapter establishes the two sides of the test but not their connection. Experience-memory studies show that retrieved records can influence later actions, while structure-sensitive exemplar selection shows that changing the retrieval representation can change which example is chosen (Li et al., 2025; Xiong et al., 2026). It remains unclear whether such a selection change carries through a multi-step interactive episode when memory contains a full trajectory and the target cookbook is already visible. The N/R/D analysis evaluates that connection directly.

Table 2.1 summarises how the study adapts, rather than exactly reproduces, the main methodological precedents. The final column identifies the specific gap addressed by the present experiment.

| Research strand            | Representative work                 | Element adapted here                                        | Remaining question tested in CookingWorld                           |
|----------------------------|-------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|
| Episodic experience memory | ExpeL; Experience-Following; ExpRAG | Frozen gold-demonstration bank and inference-time retrieval | Does a retrieved trajectory improve a complete interactive episode? |
| Dense exemplar selection   | KATE; EPR                           | Embedding-based Top-1 retrieval as a substantive baseline   | Does off-the-shelf semantic selection provide useful experience?    |



| Research strand               | Representative work | Element adapted here                                            | Remaining question tested in CookingWorld                                     |
|-------------------------------|---------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| Entity abstraction            | SGA-MCTS; STARE     | Ingredient-name replacement with anonymous role tokens          | Does suppressing entity identity change alignment or utility?                 |
| Interactive text environments | TextWorldExpress    | Reproducible environment dynamics and official terminal scoring | Can retrieval alignment and realised utility be measured on the same targets? |

**Table 2.1. Literature-to-design mapping. The listed systems motivate protocol components; none is claimed to be reproduced in full.**

## 3. Research Design and Evidential Logic

### 3.1 Temporal sequence and evidential status

The thesis reports one completed main study and two later follow-ups. The retained main-study analysis evaluates No Memory, Raw Semantic retrieval, and De-lexicalised Semantic retrieval. Only after those outcomes were unblinded was the permuted-source assignment study designed; only after that study was unblinded was the P/C active-control follow-up designed. Each later protocol was frozen before its own outcomes. The sequence therefore supports progressively narrower explanations, not one factorial decomposition planned in advance.

The revision preserves that chronology. R–N and D–R remain secondary comparisons from the original protocol, whereas R2–P and P–C are primary only within their separately frozen follow-ups.

### 3.2 Contrast logic

The four contrasts form an identification ladder. R–N adds the complete Raw Semantic memory package. D–R changes the retrieval representation and usually the selected source while preserving the raw agent-facing renderer. R2–P preserves the selected-source multiset, reuse counts, and presentation but breaks target-specific assignment. P–C preserves the target, donor, prompt shell, turn and phase counts, and approximate length while replacing the concrete trajectory with an abstract scaffold. Each step removes a narrower explanation, but estimates from different collection windows cannot be subtracted into a single causal decomposition.

| Research question                                                      | Contrast | Evidential status                                         | Intended interpretation                                             | Required claim boundary                                                                                 |
|------------------------------------------------------------------------|----------|-----------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| RQ1: Does de-lexicalisation change terminal success?                   | D–R      | Prespecified secondary; original Holm adjustment retained | Terminal-success increment associated with ingredient anonymisation | Also changes which source is selected; offline alignment is a construction-dependent manipulation check |
| RQ2: Does the complete Raw Semantic package improve success?           | R–N      | Prespecified secondary; original Holm adjustment retained | Complete Raw Semantic trajectory-and-assignment package             | Does not separate worked-example support, source content, or semantic assignment                        |
| RQ3: Does target-specific assignment add utility?                      | R2–P     | Prospectively frozen supplemental primary                 | Semantic assignment versus a distribution-matched permutation       | Local to one source vector and derangement                                                              |
| RQ4: Does concrete trajectory content add utility beyond the scaffold? | P–C      | Prospectively frozen follow-up primary                    | Concrete coherent trajectory versus paired abstract scaffold        | Not a pure replay-verification or single-feature effect                                                 |

**Table 3.1. Sequential contrasts, evidential status, and interpretation boundaries.**

### **3.3 Identification boundaries**

Terminal success is the behavioural outcome throughout. Source agreement and exact anonymised correspondence establish whether a retrieval manipulation changed selection under the study's internal representation; they do not establish usefulness or agent uptake. Likewise, descriptive failure categories can locate where episodes ended without identifying a causal mechanism. Chapter 4 specifies how each condition and estimand was implemented, while Chapters 5 and 6 keep these evidence levels separate in reporting and interpretation.

These boundaries prevent three substitutions: a package effect cannot be relabelled as a ranking effect, an internal alignment score cannot stand in for realised utility, and sequential follow-ups cannot be presented as one prospectively planned decomposition. The resulting claims remain local to the frozen target sets, source bank, prompt, and acting model.

## **4. Methodology and system implementation**

Sections 4.1–4.7 describe the completed and frozen Protocol v3.2 main experiment. Section 4.8 describes a separate supplemental experiment that was designed after the main-study unblinding but prospectively frozen before any supplemental agent outcome existed. The supplement does not revise the main experiment's hypotheses, methods, or analyses.

### **4.1 System overview**

Figure 4.1 follows one target from its cookbook specification to the final episode outcome. Offline preparation produces a frozen bank of replay-verified successes and the raw and de-lexicalised embeddings required by the retained retrievers. At test time, the target cookbook is passed either directly to the agent (N) or through one of two retrieval paths (R or D). Retrieval occurs once, before the agent takes its first action.

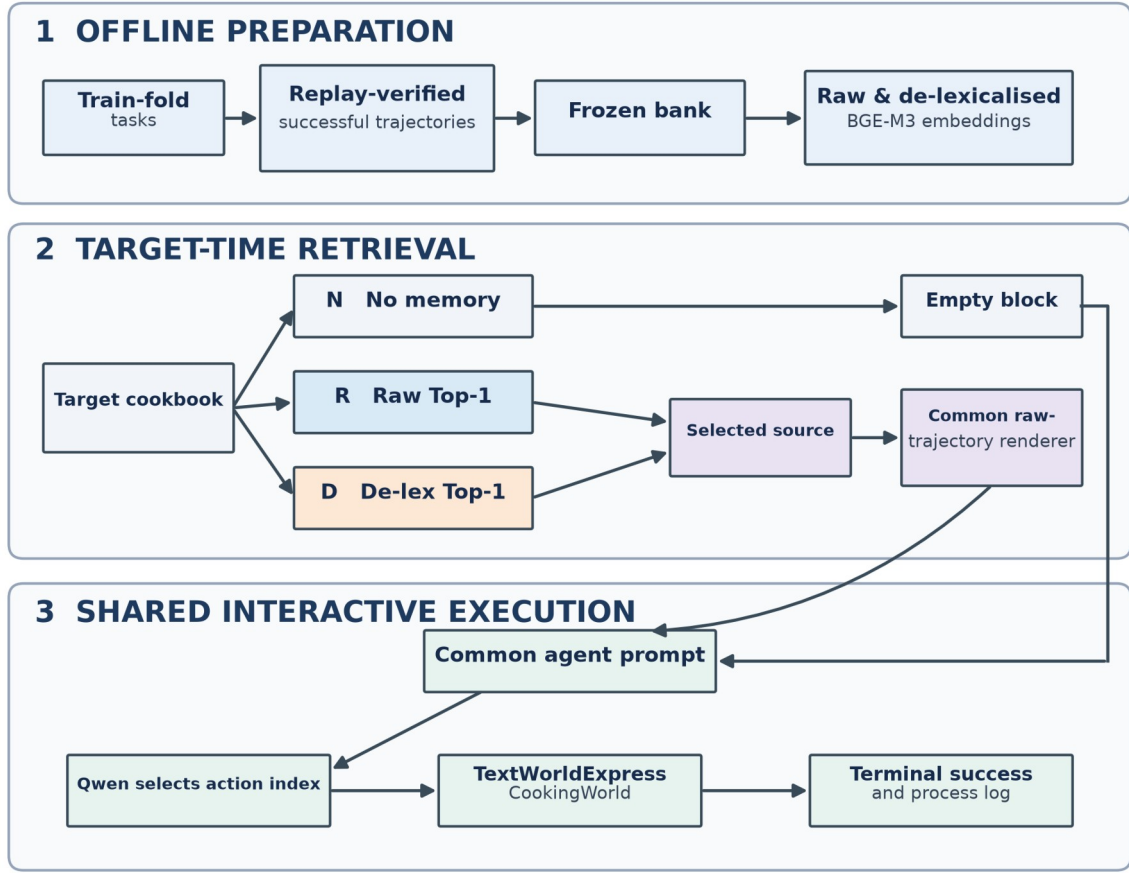


Figure 4.1. Retrieval and execution pipeline. Retrieval occurs once before interactive execution; all memory conditions use the same agent-facing trajectory renderer after source selection.

The experimental control occurs after source selection. R and D may choose different sources, but the chosen trajectory is always converted to the same agent-facing format. The target cookbook, policy model, action interface, step limit, and success criterion are identical across retained conditions.

Whether the two retrieval representations actually produced source-level separation was evaluated through the outcome-blind offline diagnostics reported in Chapter 5. The condition definitions specify ranking procedures; the realised source identities establish whether the intervention changed the prompt.

## 4.2 Environment, task input, and experience bank

Experiments use CookingWorld in TextWorldExpress 1.1.0 (Jansen & Côté, 2023). Each task contains two ingredients across three locations; doors, distractors, and inventory limits are disabled, and episodes are capped at 50 actions. This controlled configuration keeps procedural choice visible in the outcome, at the cost of generality to larger environments.

Because CookingWorld’s initial message omits the recipe, a deterministic materialiser reads the cookbook information available within the task before retrieval. The same complete cookbook is supplied in all three retained conditions without revealing locations, legal sequences, hidden state, or outcomes. Memory therefore contributes an execution example rather than otherwise unavailable recipe information.

The frozen bank archive contained 5,000 train-fold trajectories that replayed to official success. Cookbook deduplication left 3,121 indexed candidates representing 136 recipe signatures. Bank, development, pilot, formal-test, and reserve seeds are disjoint, and both the archive and index were made read-only before formal selection. Appendix A documents trajectory generation, deduplication, record contents, and replay checks.

Each indexed record contains the source cookbook, parsed ingredient–operation chains, the chronological successful action sequence, and a canonical raw rendering. The replay requirement ensures that memory-bank membership does not depend on an LLM judge or retrospective interpretation of success. Deduplication prevents identical cookbook texts from occupying multiple retrieval positions, while retaining naturally repeated procedural signatures across different source tasks.

Each stored trajectory is the environment-generated gold action sequence for its source task, re-executed and verified to reach official success. ‘Experience’ in this thesis is therefore operationalised as a replay-verified, environment-generated gold demonstration, not as a trajectory previously produced by the acting LLM; agent-generated successful trajectories may contain exploration noise absent from these materials.

### 4.3 Retrieval conditions

Table 4.1 defines the retained intervention. Both memory conditions retrieve one source from the same bank and show its canonical raw trajectory to the agent. They differ only in the text representation used for dense ranking.

| Condition                   | Selection rule                                     | Content added to the agent prompt        |
|-----------------------------|----------------------------------------------------|------------------------------------------|
| N — No Memory               | No source selected                                 | No past trajectory                       |
| R — Raw Semantic            | BGE-M3 cosine Top-1 over raw cookbooks             | Selected source in the common raw format |
| D — De-lexicalised Semantic | BGE-M3 cosine Top-1 after ingredient anonymisation | Selected source in the common raw format |

**Table 4.1. Retained retrieval conditions. R and D select from the same frozen source bank and use the same agent-facing raw-trajectory renderer.**

Only the highest-ranked source is shown, which keeps source identity unambiguous but does not test multi-source prompting. All semantic retrieval uses the same frozen BAAI/bge-m3 dense encoder (Chen et al., 2024); exact encoding settings appear in Appendix A.

R and D use the same ranking function:

$$s(q, e) = \cos(E(\tilde{x}_q), E(\tilde{x}_e)),$$

where  $q$  is the target query,  $e$  is a candidate experience,  $E$  is the frozen encoder, and the two tilded  $x$  terms are the condition-specific text renderings of  $q$  and  $e$ . R embeds raw cookbooks. D replaces ingredient names with `role_1`, `role_2`, and so forth in order of first appearance, without reordering sentences or operations. De-lexicalisation affects selection only; the selected trajectory is always rendered in the common raw format.

#### 4.4 Agent interface and episode execution

The common prompt contains the policy instruction, complete target cookbook, current observation, and a deterministically ordered list of admissible actions. N contains an empty memory block. Under R and D, the chosen source is rendered with its cookbook, observations, actions in their original order, and confirmation of source success. Retrieval scores, anonymous roles, and condition labels are never shown.

At every environment step, the agent returns a zero-based index into the admissible-action list. Restricting the output in this way removes command spelling and parsing as major sources of failure. It also narrows the claim: the experiment evaluates action choice among currently legal commands, not open-ended command generation.

| Component             | Setting                                 |
|-----------------------|-----------------------------------------|
| Environment           | TextWorldExpress 1.1.0, CookingWorld    |
| Agent model           | qwen3.7-plus-2026-05-26                 |
| Decoding              | Thinking enabled; temperature 0.0       |
| Episode limit         | 50 environment actions                  |
| Dense retriever       | Frozen local BAAI/bge-m3                |
| Memory shown to agent | One canonical raw successful trajectory |

**Table 4.2. Frozen environment, model, retrieval, and prompting configuration used in formal collection.**

An episode begins with a fresh model context. The observation and admissible actions are sent to the agent; its selected action is executed; and the next observation is returned. The loop ends at official success, another terminal state, or the 50-action limit. A nominal temperature of zero does not guarantee identical hosted-model outputs, particularly with reasoning enabled, which is why independent repetitions are retained.

Transport failures are handled under a fixed retry policy, while invalid choices and unsuccessful episodes remain scientific observations. Appendix A records the timeout, retry, and service-metadata rules used to distinguish technical failure from task failure.

## 4.5 Experimental design

The target task is the independent unit. Every formal target is analysed under N, R, and D, with two agent executions nested within each target  $\times$  condition cell. The repetitions estimate policy variability but are not counted as separate target tasks. Blinding, batch interleaving, and run-control procedures are reported in Appendix A.

The retained main-study analysis contains 675 targets and 4,050 planned N/R/D episodes. Of these, 4,044 yielded accepted records; all 675 targets retained at least one valid execution in each retained condition.

$$675 \text{ targets} \times 3 \text{ retained conditions} \times 2 \text{ runs} = 4,050 \text{ planned episodes.}$$

| Design element                        | Retained main-study specification                               |
|---------------------------------------|-----------------------------------------------------------------|
| Independent experimental unit         | Target task                                                     |
| Conditions per target                 | Three: N, R, and D                                              |
| Repetitions per target-condition cell | Two nested executions                                           |
| Formal targets                        | 675                                                             |
| Planned retained episodes             | 4,050                                                           |
| Retained contrasts                    | R–N and D–R; both prospectively specified secondary comparisons |
| Uncertainty estimate                  | 10,000-resample target-cluster bootstrap, 95% CI                |
| Sensitivity test                      | Two-sided target-level sign-flip test                           |
| Multiplicity                          | Original Holm-adjusted p-values retained                        |

**Table 4.3. Formal target-blocked experimental design and analysis specification.**

The 675-target sample size was fixed by the original protocol before the retained comparisons were analysed. Because the original power calculation targeted a different confirmatory contrast, it is not reinterpreted as prospective power for D–R or R–N. The retained analysis uses all available N/R/D data and reports effect sizes with target-cluster confidence intervals.

The pilot supplied nuisance estimates for binary success, target heterogeneity, and dependence between the two repetitions in a cell; its observed arm differences were not used to select the formal sample. Conditions were counterbalanced within target and interleaved in collection time so that a provider-side change could not align systematically with one arm. Repetition estimates hosted-policy variability, while blocking ensures that each contrast compares the same target tasks.

## 4.6 Outcomes and statistical analysis

Official terminal success is the primary outcome: the agent must prepare and consume the required recipe within 50 actions. It is produced by the environment, so the main analysis does not rely on an LLM judge.

Outcome-blind retrieval diagnostics precede online results and cover R/D source agreement, exact anonymised recipe correspondence, and source concentration. A post-unblinding mutual-information diagnostic examines whether ingredient identity predicts operation-chain labels in the generated bank. Failure categories were also operationalised after unblinding and are descriptive only. Neither process traces nor prompt hashes establish that a particular action was caused by memory.

Exact correspondence was used because the construct of interest was full recipe-level procedural correspondence rather than graded structural resemblance. CookingWorld recipes are conjunctive: successful completion requires every ingredient to satisfy its complete required transformation specification, including the prescribed order of operations. A source that matches all but one of the required chains may therefore share substantial structure but does not encode the complete target procedure.

Once the recipe representation is fixed, exact matching provides a deterministic criterion without requiring additional edit costs, feature weights, or similarity thresholds. Accordingly, a score of zero denotes failure to satisfy exact anonymised correspondence; it does not imply the absence of partial structural overlap or downstream utility. The binary criterion was used only as an outcome-blind offline diagnostic; R and D ranked candidates by continuous cosine similarity.

These evidence levels answer different questions. Source agreement establishes whether R and D produced different prompts. Exact correspondence describes the property privileged by D, but is not treated as external ground truth of utility. Failure categories can locate where unsuccessful episodes ended, yet cannot convert an outcome comparison into a causal account of how the model used memory.

Let  $Y_{i,c,r} \in \{0,1\}$  denote official terminal success for target  $i$ , condition  $c$ , and repetition  $r$ .

Let  $V_{i,c}$  denote the set of technically valid repetitions for target  $i$  under condition  $c$ , and let  $n_{i,c} = |V_{i,c}|$ . The target–condition mean is

$$\bar{Y}_{i,c} = \frac{1}{n_{i,c}} \sum_{r \in V_{i,c}} Y_{i,c,r}.$$



With two valid binary runs this score is 0, 0.5, or 1; when one run is technically missing, the single valid run is the available-run mean. A target is omitted from a contrast only if either contrasted condition has no valid repetition. No retained main-study target met that exclusion rule. For each contrast A–B, the reported effect is the mean of target-level differences  $Y_{i,A} - Y_{i,B}$ .

$$\Delta_i = \bar{Y}_{i,A} - \bar{Y}_{i,B}.$$

Averaging does not create additional observations; it summarises the two stochastic executions so that each target receives equal weight. A 10,000-resample target-cluster bootstrap supplies the 95% confidence interval, and a two-sided target-level sign-flip test supplies a sensitivity p-value. The original Holm-adjusted p-values are retained to avoid post hoc loosening of multiplicity control.

R–N and D–R were members of the original Holm-corrected secondary family; the thesis retains their originally adjusted p-values. The failure and operation-overlap analyses remain descriptive only.

The target-level analysis was chosen because the intervention is assigned within target and the two calls in a cell are stochastic repetitions. Cluster resampling preserves that dependence and gives every target equal influence, including cells with one technical missing run. The sign-flip test evaluates the paired null without treating episode-level calls as independent trials.

#### **4.7 Quality assurance and scope**

Every bank trajectory replayed to official success; experimental partitions were disjoint; and preprocessing, ranking, and rendering were deterministic. Excluded material supported parser, prompt, scorer, reliability, and cost checks. Retrieval outputs, the run manifest, and analysis code were frozen before blinded online collection. Appendix A records these safeguards and the outcome-blind amendments.

The retained analysis identifies only Top-1, frozen-bank, CookingWorld-specific package and representation contrasts under one acting model. It neither tests dynamic memory or multi-source prompting nor supports causal claims about structural correspondence, universal retrieval performance, or human-like structure mapping.

Blinding was directed at condition-level outcomes rather than at the deterministic retrieval files, which had to be inspected for gate acceptance. Before unmasking success, technical acceptance checked planned-cell accounting, prompt and source bindings, official-scorer reconstruction, retry compliance, and unresolved missingness. These checks protect against selective rerunning or misbinding while leaving scientific interpretation until after the dataset was closed.

## 4.8 Prospectively frozen supplemental permuted-source experiment

### 4.8.1 Motivation and temporal status

After v3.2 was completed and unblinded, a supplemental experiment was designed to distinguish target-specific semantic assignment from the value retained when that assignment is broken. Its protocol was frozen before any supplemental outcomes existed and does not alter the completed main study. The control permutes semantically retrieved sources rather than sampling uniformly from the bank, so it is a **permuted-source baseline**, not generic random retrieval.

### 4.8.2 Conditions and source assignment

The supplement contains three conditions. N2 provides no retrieved source. R2 uses the frozen Raw Semantic Top-1 retriever to assign one replay-verified successful trajectory to each supplemental target. P permutes the complete R2 source-assignment vector across the same targets.

| Condition | Source assignment                            | Agent-facing content                                                  |
|-----------|----------------------------------------------|-----------------------------------------------------------------------|
| N2        | No source                                    | Empty experience wrapper                                              |
| P         | Permutation of the complete R2 source vector | Permuted successful trajectory in the common raw format               |
| R2        | Frozen Raw Semantic Top-1 retriever          | Semantically retrieved successful trajectory in the common raw format |

**Table 4.4. Conditions in the prospectively frozen supplemental permuted-source experiment.**

The permutation preserves the complete source multiset and reuse counts while requiring  $P_i \neq R2_i$  for every target and a different donor target. It does not inspect overlap, correspondence, difficulty, or outcomes. P and R2 then use identical rendering and prompt placement, isolating target-to-source assignment while holding the source distribution fixed.

### 4.8.3 Design, masking, and execution

The supplement uses the 675 completed main-study targets plus the first 125 targets in the frozen reserve order. It is prospective for its new outcomes but is not an independent target-sample replication. Every target appears in all three conditions with two executions per cell:

$$800 \text{ targets} \times 3 \text{ conditions} \times 2 \text{ repetitions} = 4,800 \text{ formal episodes.}$$

Twelve excluded smoke episodes verified binding, rendering, scoring, and service operation. Frozen reverse-balanced sequences, masked labels, and interleaved batches protected the comparison; the target remained the inferential unit. Operational amendments and the technical ledger are reported in Appendix A.9.

#### 4.8.4 Estimands, power, and analysis

The primary estimand is the target-level average difference between Raw Semantic and permuted assignment:

$$\tau_{R2-P} = \frac{1}{800} \sum_{i=1}^{800} (\bar{Y}_{i,R2} - \bar{Y}_{i,P}).$$

This asks whether semantic assignment adds utility relative to a permutation of the same trajectories. Target means follow the available-run rule in Section 4.6. The frozen analysis combines a two-sided paired target-level difference test with two one-sided tests (TOST) of equivalence within  $\pm 5$  percentage points; estimates and confidence intervals are reported regardless of either decision. The reported 95% effect interval uses a 10,000-resample target-cluster bootstrap, whereas the 90% equivalence interval is the t-based TOST interval calculated from the target-level paired-difference mean and standard error.

The five-percentage-point margin was a study-specific practical boundary chosen after the main-study outcomes were known but fixed before any supplemental outcome was inspected. It is approximately one quarter of the main-study R–N package difference of 20.59 percentage points. In this low-stakes environment, an assignment increment below that magnitude was judged insufficient to change the engineering decision about whether target-specific retrieval was worth deploying, and five points was also commensurate with the smallest effect that the two-repetition design could detect with reasonable stability. The boundary is not presented as a universal threshold for agent deployment.

The reported confidence intervals condition on the single realised derangement; they do not include sampling variability across the space of admissible permutations. Conclusions about assignment therefore apply to this realised mapping, and ‘permuted assignment’ should not be read as generic random retrieval.

Two Holm-corrected secondary contrasts compare P and R2 with concurrent No Memory. Because 675 of 800 targets overlap the main study, R2–N2 is a cross-window stability check, not an independent replication.

Under the most conservative frozen scenario, estimated difference-test power was 79.72% and TOST power under a true zero effect was approximately 76.4%, both below the original 0.80 gate. Before any outcome was inspected, a dated amendment approved proceeding at the fixed 800-target resource ceiling without changing the estimand or analysis. The full power grid and decision record appear in Appendix A.9.

The equivalence analysis is important because a non-significant difference alone would not show that semantic assignment was practically negligible. TOST instead asks whether effects at or beyond either five-point boundary are inconsistent with the data. The supplement therefore complements rather than repairs the main study: it addresses the assignment question prospectively, but was motivated after the main result and retains that temporal boundary.

## **4.9 Prospectively frozen P/C protocol-control follow-up**

### **4.9.1 Motivation and temporal status**

After the main study and the permuted-source supplement were unblinded, a second follow-up was designed to address the unresolved contrast between a concrete trajectory and any non-empty worked interaction scaffold. Its protocol, paired materials, target manifest, bindings, analysis code, and stopping rules were frozen before any P/C outcome was inspected. The experiment reused the 800 targets and P source assignments from the first supplement, so it is prospective for its own outcomes but not an independent target-sample replication.

### **4.9.2 Conditions and scaffold construction**

P presents the replay-verified successful trajectory assigned by the previously frozen non-target-specific permutation. In a common neutral shell, it includes the source cookbook, initial observation, chronological action–observation turns, and a terminal-success marker, but not a source identifier or seed. C is generated deterministically from the same P source. It preserves the renderer field order, number of turns, action–observation alternation, coarse phase sequence, terminal-success marker, and approximate length, while removing the source cookbook and recipe, concrete entities, locations, devices, operations, original commands, and original observations.

Agent-visible text never names either condition or describes an item as retrieved, permuted, replay-verified, a control, or a sham. Neither P nor C displays source admissible-action lists or selected zero-based indices; the live zero-based response rule is identical across arms. Static QA covered 515 distinct P-source trajectories, each paired with its deterministic C scaffold and reused across the 800 target assignments. Common renderer and full prompt shells matched exactly after substituting the experience body; turn and phase counts matched; lexical leakage checks found no source- or target-specific content in C. The C-to-P word-count ratio ranged from 0.951 to 1.006, and the local Qwen3-family proxy-token ratio ranged from 0.900 to 0.915. The exact hosted tokenizer was not claimed. A complete unedited paired example is reproduced in Appendix A.11.1.

### **4.9.3 Design, masking, and execution**

Every target received P and C twice, yielding  $800 \text{ targets} \times 2 \text{ conditions} \times 2 \text{ repetitions} = 3,200$  formal executions (1,600 target–condition cells). The paired arms shared the same frozen P donor. A seeded run order interleaved masked cells, while the acting model, target cookbook, environment, prompt shell, admissible-action interface, history window, repair rule, official scorer, and 50-step limit remained fixed. Eight outcome-excluded smoke executions were technically accepted before formal

collection. Immutable records were never overwritten, and claims without a recoverable provider record remained missing rather than being rerun after scientific outcomes were available.

#### 4.9.4 Estimand and analysis

For target  $i$ , valid repetitions were averaged within condition and the primary estimand was the equally weighted target-level mean of  $Y_{i,P} - Y_{i,C}$ . P–C estimates the incremental effect of supplying a concrete, coherent replay-verified trajectory rather than this particular abstract format-, length-, and phase-matched scaffold under the same frozen non-target-specific assignment. It does not identify a pure experience effect, the causal effect of replay verification, or the contribution of any single concrete trajectory feature.

The frozen report includes a 10,000-resample paired target bootstrap 95% confidence interval, a two-sided Monte Carlo sign-flip sensitivity test, and paired-difference TOST against a prospectively fixed  $\pm 5$ -percentage-point margin. The 90% equivalence interval is t-based and uses the target-level paired-difference mean and standard error; it is therefore distinct from the bootstrap 95% effect interval. The sign-flip calculation is treated as a paired-symmetry sensitivity rather than exact design-based randomisation inference. A target is omitted only if either arm has no valid repetition. Extreme binary imputations of missing executions and a paired t-test were reserved as transparent, unplanned robustness checks. At the fixed 800-target resource ceiling, conservative prospective simulations estimated 79.0% sign-flip power for a five-point difference and 76.7% TOST power for equivalence under a true zero effect.

## 5. Results

This chapter reports the retained N/R/D main-study analysis followed by two separately frozen follow-ups. The first tests target-specific assignment through a distribution-matched permutation; the second tests concrete trajectory content against a paired abstract worked-protocol scaffold. Main-study contrasts retain their original secondary evidential status.

| Tier                          | Analyses                                                 | Status                                                                        |
|-------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------|
| Retained main-study contrasts | R–N and D–R                                              | Prospectively specified secondary analyses; original Holm adjustment retained |
| Predeclared diagnostics       | R/D source agreement and exact anonymised correspondence | Frozen outcome-blind                                                          |
| Post-unblinding exploratory   | Failure taxonomy and operation-overlap stratification    | Descriptive only                                                              |
| Prospective follow-ups        | R2–P assignment control; P–C active control              | Frozen before their own outcomes                                              |

**Table 5.1. Evidence tiers used in this chapter.**

## 5.1 Data completeness and integrity

Of 4,050 planned N/R/D episodes, 4,044 yielded accepted records and six (0.15%) were technical missingness. Effective counts were 1,347 (N), 1,348 (R), and 1,349 (D); every retained contrast included all 675 targets through the available-run rule. No record was excluded for scientific reasons.

Every target–condition cell retained at least one valid execution, so missingness never removed a target from a contrast. Prompt bindings, selected-source identifiers, and reconstructed official scores passed technical acceptance before condition outcomes were unmasked. The small imbalance in episode counts therefore changes only whether six cell means use one rather than two repetitions.

## 5.2 Offline retrieval behaviour

Retrieval selections and core diagnostics were frozen before formal agent calls. D and R selected different sources for 673 of 675 targets (99.7%). The de-lexicalised intervention therefore changed the selected trajectory on almost every target; the concern is not that the intervention failed to occur, but that its headline alignment measure is representation-dependent.

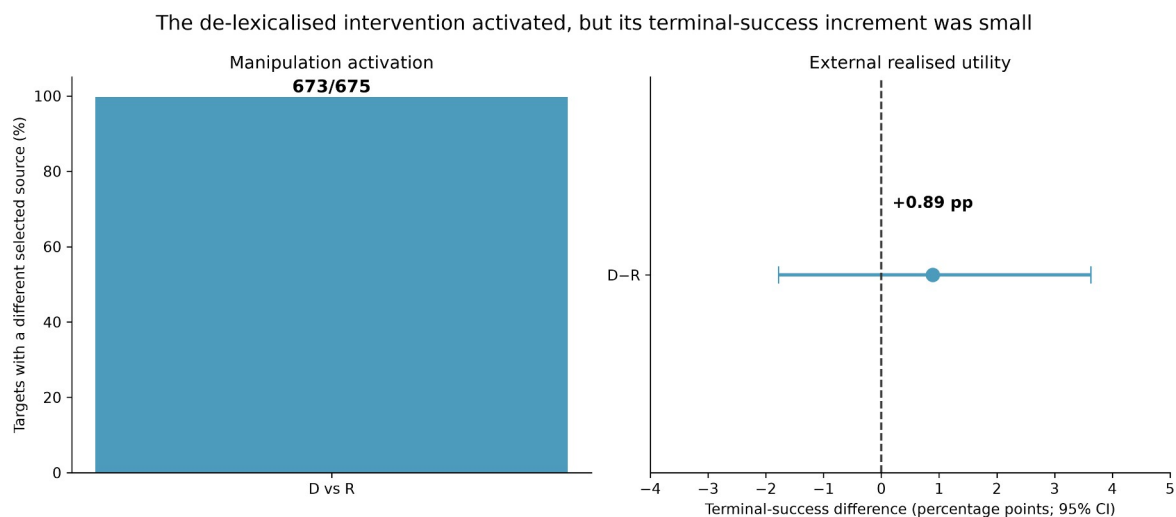
Exact anonymised recipe correspondence rose descriptively from 44 of 675 targets (6.52%) under R to 675 of 675 (100%) under D, a +93.48-percentage-point manipulation check. No inferential claim is attached to this construction-dependent statistic. D retrieves from ingredient-anonymised cookbook text, while exact correspondence is computed from ingredient-anonymised ordered operation chains; the measure is therefore not independent of the intervention. The increase documents the intended behaviour of the representation, but it is not external validation that D found objectively better experiences and does not establish usefulness or agent uptake.

A post-unblinding bank diagnostic found normalised mutual information of 0.371 between ingredient identity and ordered operation-chain labels. Ingredient words therefore carried procedural information in these generated materials, although the diagnostic does not identify their causal retrieval value.

Taken together, the offline results show a real source-selection intervention but a circular headline alignment diagnostic. Between R and D, selections changed on 673 of 675 targets. Exact correspondence then became universal under the same abstraction that defined D retrieval. The central question is therefore whether saturating this representation-specific metric was accompanied by a change in terminal behaviour.

Figure 5.1 separates intervention activation from external outcome. De-lexicalisation changed the selected source on 673 of 675 targets, confirming that the manipulation altered the prompt. Its

terminal-success increment over Raw Semantic retrieval was +0.89 points with a confidence interval spanning modest benefit and harm. The exact alignment statistic reported above is therefore treated only as an internal manipulation check.



**Figure 5.1. Manipulation activation and realised utility.** De-lexicalised retrieval selected a different source from Raw Semantic retrieval on 673 of 675 targets (99.7%). The corresponding D–R terminal-success estimate was +0.89 percentage points (95% target-cluster bootstrap CI [–1.78, +3.63]). Source change confirms intervention activation; it does not establish that the selected source was better or used by the agent.

### 5.3 Retained main-study results

Table 5.2 reports target-level mean success by condition. For each target, the two binary executions in a condition were averaged to a score of 0, 0.5, or 1, so that each target contributes equally.

| Condition                   | Target-level mean success |
|-----------------------------|---------------------------|
| N — No Memory               | 63.33%                    |
| R — Raw Semantic            | 83.93%                    |
| D — De-lexicalised Semantic | 84.81%                    |

**Table 5.2. Terminal success by condition (675 targets per condition).** Displayed percentages are rounded to two decimals; contrasts were calculated from unrounded target-level values.

### 5.4 Main-study contrasts

Table 5.3 reports the two retained main-study contrasts with their original confidence intervals and Holm-adjusted p-values. These were prospectively specified secondary analyses, not the protocol's sole confirmatory comparison.

| Contrast | Difference | 95% CI           | p (Holm) | Status                 |
|----------|------------|------------------|----------|------------------------|
| R–N      | +20.59 pp  | [+16.89, +24.22] | < .001   | Prespecified secondary |
| D–R      | +0.89 pp   | [–1.78, +3.63]   | .883     | Prespecified secondary |

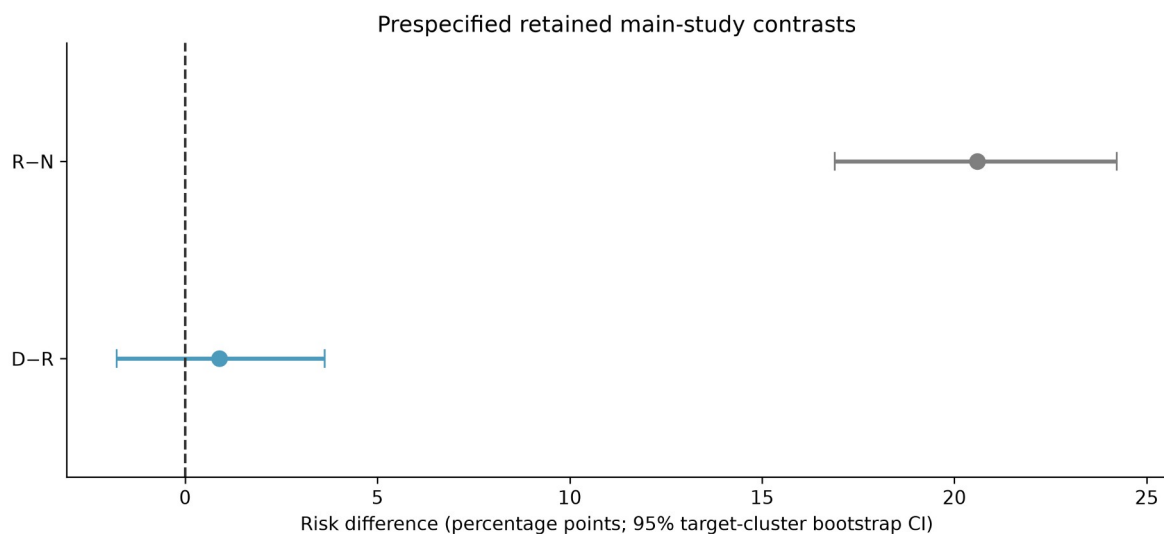
**Table 5.3. Retained main-study target-level contrasts.** Confidence intervals use the frozen target-cluster bootstrap; p-values preserve the original Holm adjustment.



R–N shows a large memory-package benefit despite complete cookbook visibility; it does not establish episode-level uptake or isolate which trajectory property caused the change. D–R shows that the large representation-dependent correspondence gain was accompanied by only a small estimated success difference, with a confidence interval spanning modest benefit and harm. This supports a bounded alignment–utility gap but not formal equivalence or a claim of exactly zero effect.

This pattern is consistent with a positive average behavioural effect of adding the non-empty Raw Semantic package: success under R was substantially higher than under N. At the same time, the retained analysis gives no evidence that removing ingredient identity added a substantial average utility increment beyond raw semantic access.

The wording “memory-package benefit” is deliberate. R–N jointly changes the presence of a replay-verified trajectory, its raw rendering, and its semantic assignment. Without an assignment-preserving baseline in the main study, the contrast cannot determine how much of the improvement belongs to semantic relevance rather than to receiving a worked episode. That unresolved decomposition motivated the separate supplement.



**Figure 5.2. Retained target-level terminal-success contrasts.** Points show risk differences in percentage points and horizontal bars show 95% target-cluster bootstrap confidence intervals. R–N and D–R were prospectively specified secondary contrasts; their originally Holm-adjusted p-values are reported in Table 5.3.

## 5.5 Failure analysis

The executable taxonomy was implemented after unblinding and is descriptive. Across the 915 failed episodes retained from N, R, and D, 804 (87.87%) were ingredient–operation errors, 99 (10.82%) were premature-consumption errors, and 12 (1.31%) were navigation or step-limit failures. From N to R, roughly 74% of the failure-rate reduction was associated with fewer ingredient–operation errors and 24% with less premature consumption. Under D, 202 of 205 failures were ingredient–operation

errors despite exact anonymised source correspondence. These patterns are consistent with an execution bottleneck but do not establish one causally.

Navigation contributed little to either the residual error pool or the memory-associated improvement. The principal descriptive change was fewer irreversible transformations inconsistent with the visible recipe, followed by fewer episodes that consumed an ingredient prematurely. Exact correspondence did not eliminate these errors, suggesting that access to an applicable source and successful application remain distinct stages.

At the episode level, ingredient–operation failures fell from 29.99% under N to 14.69% under R, while premature consumption fell from 6.09% to 1.04%. Navigation or step-limit failures changed by only 0.30 points. These unadjusted descriptive rates support an execution-scaffolding interpretation more directly than a navigation-scaffolding account.

## **5.6 Exploratory operation-overlap analysis**

Post-hoc stratification found positive R–N point estimates both when targets required no transformation (+15.5 pp) and when source and target shared no operations (+4.8 pp). Estimates were larger with partial (+25.1 pp) or full (+23.2 pp) overlap, with no visible dose response beyond coarse overlap. Because overlap was not assigned and baseline difficulty differed sharply across strata, this pattern is descriptive and compatible with both generic episode organisation and coarse operation scaffolding. The complete table and caveats appear in Appendix A.10.

The no-transformation subgroup is informative because its benefit cannot arise from copying a transformation sequence. Conversely, larger estimates under positive overlap suggest that demonstrated operations may still matter. The high No-Memory baseline in the zero-overlap stratum and the non-random composition of all strata prevent a causal comparison of these components.

## **5.7 Supplemental permuted-source experiment**

This supplement was designed after the retained main-study analysis was unblinded but frozen before its own agent outcomes. It provides separate prospective evidence and does not change the evidential status of R–N or D–R.

Its purpose is narrower than an independent replication. The main study could not distinguish the value of receiving a trajectory from the additional value of assigning that trajectory through semantic retrieval. The supplement preserves the complete Raw Semantic source distribution while breaking only its target assignment, thereby addressing that decomposition prospectively.

### 5.7.1 Completeness, integrity, and a pre-analysis deviation

Of 4,800 planned episodes, 4,786 (99.71%) were accepted and 14 were technical missingness: six in P, four in R2, and four in N2. No record was excluded for scientific reasons or rerun, and all 800 targets remained in every contrast. A pre-analysis schema check prematurely displayed the condition map but no outcome or aggregate result; the analysis code was unchanged. Hashes, the cross-file audit, and the dated deviation are recorded in Appendix A.9.

### 5.7.2 Assignment diagnostics

P and R2 used the same source multiset and reuse counts, differed on every target, and never assigned a donor source to its own target. Exact correspondence was 7.0% under R2 and 1.5% under P; these diagnostics played no role in assignment or acceptance.

The low incidental match rate under P confirms that permutation usually broke both semantic assignment and the study-defined structural relation. Because every P source nevertheless came from the same replay-verified vector as R2, outcome differences cannot be attributed to unequal source-frequency distributions or to one condition drawing a generally different class of stored trajectories.

This control is stronger for the assignment claim than uniform random sampling would have been. Uniform sampling could change which sources appear and how often, confounding target matching with source quality or popularity. The implemented derangement instead preserves the empirical source mixture produced by Raw Semantic retrieval and asks only whether returning each source to its original target improves outcomes.

### 5.7.3 Primary result

Table 5.4 reports target-level mean success by supplemental condition, computed as in the main study by averaging the valid executions in each target–condition cell.

| Condition                      | Target-level mean success |
|--------------------------------|---------------------------|
| N2 — No Memory (concurrent)    | 61.38%                    |
| P — Permuted source            | 82.69%                    |
| R2 — Raw Semantic (concurrent) | 84.06%                    |

**Table 5.4. Terminal success by supplemental condition (800 targets per condition). Displayed percentages are rounded to two decimals; contrasts were calculated from unrounded target-level values.**

The primary supplemental estimand is the target-level difference between semantic and permuted assignment:

$$\hat{\tau}_{R2-P} = +1.375 \text{ pp}, \quad 95\% \text{ CI } [-1.25, +4.06], \quad p = .339,$$

with the confidence interval from the frozen target-cluster bootstrap and the p-value from the two-sided target-level sign-flip test. The two-sided test provides no evidence that target-specific semantic assignment outperformed the permuted assignment of the same trajectories.

The prospectively specified and frozen equivalence procedure supports a stronger, margin-specific conclusion. Both one-sided tests at the frozen  $\pm 5$ -percentage-point margin rejected, and the t-based 90% confidence interval  $[-0.84, +3.59]$  lies entirely within those bounds. The target-specific and permuted conditions therefore met the prospectively specified, study-specific  $\pm 5$ -point equivalence criterion. This does not establish a zero effect: the point estimate is positive, and the bootstrap 95% interval remains compatible with a small benefit of up to approximately four percentage points.

As an unplanned sensitivity, the same 90% interval  $[-0.84, +3.59]$  would also satisfy a  $\pm 4$ -point margin but not a  $\pm 3$ -point margin; the equivalence conclusion is therefore specific to margins of approximately four points or wider.

The difference test and equivalence test answer complementary questions. Failure to reject zero alone would leave large positive or negative effects possible; the TOST result instead excludes effects at or beyond the frozen practical bounds under the specified analysis. Its interpretation remains local to this source vector, permutation, model, and target distribution.

A post-unblinding binary worst-case sensitivity kept the primary-arm 90% intervals within the  $\pm 5$ -point bounds (Appendix A.9).

#### 5.7.4 Secondary contrasts and cross-window stability

The two secondary contrasts form one Holm-corrected family (Table 5.5).

| Contrast       | Difference | 95% CI             | p                  | Status                                                |
|----------------|------------|--------------------|--------------------|-------------------------------------------------------|
| R2-P (primary) | +1.375 pp  | $[-1.25, +4.06]$   | .339 (uncorrected) | Supplemental primary; equivalent within $\pm 5$ pp    |
| P-N2           | +21.31 pp  | $[+18.19, +24.31]$ | < .001             | Supplemental secondary                                |
| R2-N2          | +22.69 pp  | $[+19.31, +25.94]$ | < .001             | Supplemental secondary; concurrent stability estimate |

**Table 5.5. Supplemental primary and secondary target-level contrasts. The primary p-value is from the frozen sign-flip test and is not part of the corrected family.**

The permuted package improved success by +21.31 points over concurrent No Memory and retained approximately 94% of the concurrent supplemental R2-N2 package effect by point estimate (21.31/22.69). This ratio is descriptive, not a causal proportion. R2-N2 showed numerical stability relative to the main collection on a largely overlapping target set; it is not an independent replication.

Thus the supplement supports the claim that most of the observed Raw Semantic package benefit was retained when target-specific assignment was broken. It does not show that replay verification or

successful source outcomes caused that benefit, because no unsuccessful-trajectory condition was included. Nor does the numerical stability across windows supply independent target-sample evidence: 675 of the 800 supplemental targets were reused.

#### **5.7.5 Failure profile**

In the post-unblinding descriptive taxonomy, ingredient–operation errors accounted for 88.85% of supplemental failures. Counts are reported in Appendix A.9; they do not establish source-caused failure.

#### **5.7.6 Summary**

Together, the studies separate a large memory-package effect retained under permutation, a small semantic-assignment increment equivalent within  $\pm 5$  points, and a small D–R estimate despite saturation of the representation-dependent alignment metric. These are package, assignment, and representation contrasts, not causal estimates of structural correspondence or source success.

### **5.8 P/C protocol-control follow-up**

This second follow-up was designed after the preceding studies were unblinded but frozen before its own outcomes. It directly tests whether the benefit retained under permutation can be reproduced by an abstract worked-protocol scaffold that controls the non-empty block, chronology, phase structure, approximate length, and successful ending.

#### **5.8.1 Completeness, integrity, and deviations**

Of 3,200 formal executions (1,600 target–condition cells), 3,192 executions produced accepted immutable records: 1,596 in P and 1,596 in C. Eight executions—four per arm—were consumed without a recoverable record after two outcome-blind stale-run incidents and were never rerun. Every affected target retained its other repetition in that arm, so all 800 targets remained evaluable under the frozen available-run rule. Technical acceptance found no model, prompt, binding, scorer, leakage, or immutable-record failure.

Formal collection initially used five workers under an explicit concurrency amendment and stopped at its technical gate after 68 accepted records. Outcome-blind amendments repaired ledger reconciliation and resumed collection with four workers, a durable launcher, and a stop-only watchdog. Across the final resume epoch, 56 HTTP 429 responses affected 52 accepted records but resolved within the frozen retry rules and did not meet the prospective stopping threshold at acceptance. No condition outcome was inspected before technical closure, and the scientific materials and analysis hashes remained unchanged.

### 5.8.2 Primary result

Table 5.6 reports equally weighted target-level success, averaging available valid repetitions within each target–condition cell.

| Condition                             | Target-level mean success |
|---------------------------------------|---------------------------|
| C — Abstract worked-protocol scaffold | 60.94%                    |
| P — Concrete permuted trajectory      | 81.00%                    |

**Table 5.6. Terminal success in the P/C protocol-control follow-up (800 targets per condition).**

The target-level P–C difference was +20.06 percentage points, with a paired target-bootstrap 95% confidence interval of [+17.13, +23.06]. The two-sided Monte Carlo sign-flip sensitivity gave  $p < .001$ . The 90% paired-difference interval was [+17.53, +22.60], entirely above the prospectively specified +5-point boundary. The result therefore supports a practically meaningful advantage for P over C rather than equivalence within  $\pm 5$  points.

An unplanned paired t-test at the same target level gave  $t(799) = 13.029$ ,  $p < .001$ . This robustness calculation is consistent with the frozen estimate but does not replace the target bootstrap, sign-flip sensitivity, or prospective TOST decision.

### 5.8.3 Missingness sensitivity and interpretation

Extreme binary imputations could not explain the result. Setting every missing P execution to failure and every missing C execution to success yielded +19.81 points; reversing those assignments yielded +20.31 points. The primary available-repetition estimate of +20.06 points lies between these bounds.

P–C directly rejects the specific explanation that the benefit retained under permutation can be fully reproduced by this abstract scaffold alone. It does not identify replay verification as causal: P also differs from C in concrete commands, observations, source-domain information, natural-language coherence, and actionability. C's success rate (60.94%) was numerically close to N2 in the earlier collection (61.38%), but that cross-window comparison is descriptive only and does not establish that abstract scaffolds are generally equivalent to empty memory. Because the experiments occurred in different collection windows and no concurrent N arm was included here, P–C (+20.06 points) and P–N2 (+21.31 points) must not be subtracted to infer C–N or combined into a causal decomposition.

## 5.9 Summary of results

The retained main-study analysis found a large Raw Semantic memory-package benefit and only a small De-lexicalised-versus-Raw success difference despite saturation of the study-defined alignment metric. The first follow-up showed that most of the Raw Semantic package benefit was retained when target-specific assignment was broken and that the semantic-assignment increment was equivalent

within  $\pm 5$  points. The P/C follow-up then showed that a concrete coherent trajectory outperformed a paired abstract worked-protocol scaffold by +20.06 points.

Accordingly, RQ1 supports a bounded representation-sensitive alignment–utility gap, while RQ2 concerns the complete Raw Semantic package rather than semantic matching alone. The assignment control narrows the role of target-specific matching, and P–C shows that the implemented abstract protocol scaffold cannot account for the concrete trajectory package benefit. Neither follow-up identifies a pure replay-verification effect.

## 6. Discussion

### 6.1 Principal findings

The strongest supported finding is that the tested memory package was useful even when target-specific matching contributed little. Raw Semantic retrieval improved terminal success by 20.59 percentage points over an empty block. In the separately frozen assignment follow-up, permuting the same successful-source vector retained most of that benefit; the estimated target-specific assignment difference was +1.375 points (95% CI [-1.25, +4.06]) and met the  $\pm 5$ -point equivalence criterion. The evidence therefore places a small practical bound on the assignment increment in this realised source vector rather than attributing the package benefit to semantic matching.

The active-control follow-up located a further component of that package. The concrete permuted trajectory produced 81.00% target-weighted success and the paired abstract scaffold produced 60.94%, a P-C difference of +20.06 points (95% CI [+17.13, +23.06]). This rejects the specific explanation that the benefit retained under permutation can be reproduced by this abstract scaffold alone. It does not isolate replay verification, source success, or any single content feature because P also adds concrete commands, grounded observations, domain information, natural-language coherence, and actionability. C was numerically close to the earlier N2 rate (61.38%), but the conditions were not concurrent; this observation cannot be used to subtract an abstract-scaffold effect or to generalise from C to all possible scaffolds.

The main D manipulation provides a complementary boundary result. De-lexicalisation changed 673 of 675 sources and saturated the representation-internal alignment criterion as designed, yet D differed from R by only +0.89 percentage points in terminal success (95% CI [-1.78, +3.63]). This was a prespecified secondary comparison, not a formal equivalence test. Taken together, the sequential studies support an evidence-matched conclusion: in this high-redundancy CookingWorld bank, changing selected-source correspondence from 6.52% under R to 100% under D added little utility, whereas concrete coherent trajectory content carried substantial utility relative to the implemented scaffold. The studies narrow the package effect without forming one prospectively planned arithmetic decomposition.

### 6.2 Why matching added little beyond trajectory access

Every condition received the complete target cookbook, so exact source correspondence added no hidden recipe information. A past trajectory could still demonstrate legal command use, episode



organisation, and common operations, but those functions need not require an exact match between ingredient roles and operation chains. This interpretation fits both the main and supplemental experiments: correcting explicit mismatches did not improve average success, and permuting the semantic source assignment retained most of the memory-package benefit.

Cookbook visibility changes the function of memory: a concrete source shows how a recipe becomes grounded commands and feedback rather than revealing the target recipe. P/C rules out the strongest generic-protocol account for the implemented control because abstract phase order, approximate length, and successful stopping did not reproduce P's utility. Read with the failure taxonomy, the most plausible bounded interpretation is that omitted support lay mainly in operation execution and episode termination—action forms, observed consequences, and transformation, preparation, and consumption timing. This remains interpretive because P and C also differ in grounded information, linguistic coherence, and actionability.

The descriptive failure decomposition narrows this account. From N to R, ingredient–operation errors fell by 15.30 percentage points and premature-consumption failures by 5.05 points, but navigation or step-limit failures by only 0.30 points. Positive differences on zero-operation-overlap and no-transformation targets remain consistent with episode-organisation support; larger differences with any operation overlap are consistent with operation-execution scaffolding, with no visible dose response from partial to full overlap. A primarily navigational account is therefore unlikely. These post-unblinding, non-randomised patterns support interpretation, not causal mediation, and do not separate command syntax from feedback or linguistic coherence.

Remaining failures also point toward execution, but only descriptively: 202 of 205 D failures involved an ingredient–operation error although all 675 selected sources met the exact criterion. A match did not guarantee execution through irreversible actions. With the target cookbook visible, exact assignment therefore offered little measurable value beyond a trajectory's concrete execution support.

The first supplement strengthens the assignment claim because it manipulates source allocation rather than selecting a convenient mismatch subset after outcomes are known. The P/C follow-up strengthens the content claim by holding the frozen donor, target, common shell, chronology, and broad phase sequence fixed while replacing the concrete trajectory with an abstract scaffold. Even so, neither contrast is a mechanism test. P jointly contains concrete command syntax, grounded observations, domain information, coherence, and a replay-verified successful ending, so the observed increment cannot be assigned to any one feature or to replay verification itself.

### 6.3 Retrospective design diagnosis and component activation

Bank-level signature redundancy must be separated from selected-source correspondence. By the offline-preparation stage, 3,121 bank entries collapsed to 136 recipe signatures, showing that exact matching candidates were generally available; this did not show that R already selected them. R met the exact anonymised-correspondence criterion on only 44 of 675 targets (6.52%), whereas D met it on all 675 and changed the selected source on 673 targets. D–R was therefore an activated comparison between an overwhelmingly non-corresponding raw selection and a criterion-saturating de-lexicalised selection. Its +0.89-point terminal-success difference is the retrieval–utility gap, not a comparison between two retrievers already operating within the same saturated set.

These facts motivate a component-activation gate for future retrieval studies. Before paid outcome collection, each claimed component should be required to change the final source or prompt on a prospectively specified proportion of targets, with the threshold linked to the planned effect size and power calculation; it should also demonstrate headroom on the metric it is intended to improve. The retained D–R comparison is informative precisely because D activated the intended selection manipulation: exact correspondence rose from 44 of 675 to 675 of 675 targets and 673 final sources changed, yet terminal success changed little. It does not establish that structural retrieval is generally ineffective.

### 6.4 Relation to prior work

Classic analogical-transfer research separates access to a precedent from its application (Gick & Holyoak, 1983). Here, retrieval guaranteed access by placing a source in the prompt, yet corresponding sources did not ensure correct action sequences. Holyoak and Koh (1987) also treat surface similarity as a potentially functional access cue. In this bank, ingredient identity carried information about operation chains; removing it radically changed correspondence but not terminal performance. Surface information was therefore neither pure noise nor a decisive source of realised utility.

The present setting differs from spontaneous human analogy in an important way. The source is supplied explicitly and the target relation is written in the cookbook, so the experiment does not test whether the model independently notices a distant analogy. Instead, it tests whether one algorithmic access policy provides a more useful demonstration to an acting model. This narrower framing avoids treating a domain-specific symbolic retriever as evidence of human-like structure mapping.

Experience-Following warns that similar memories may propagate misaligned behaviour (Xiong et al., 2026). CookingWorld supplies a boundary condition: even experimentally assigned mismatched sources remained beneficial on average when the correct cookbook was visible and all bank trajectories were replay-verified successes. SGA-MCTS and STARE motivate abstraction or structural alignment in exemplar selection (Xie et al., 2026; Li et al., 2025), but the present results show why an offline alignment gain still requires downstream validation.

This does not contradict those methods’ original domains. Structure-sensitive selection may be valuable when an exemplar supplies latent reasoning steps, the target specification is incomplete, or the acting policy cannot reconstruct the relevant relation from raw text. CookingWorld instead exposes the target recipe and uses a small signature space. The study contributes a cross-domain boundary test: a retrieval representation can improve its intended offline criterion without a detectable corresponding gain in terminal success.

## **6.5 Implications for retrieval evaluation**

Three checks should accompany claims about experience retrieval. First, report alignment and terminal utility separately: a domain-defined match criterion is inspectable but not external ground truth of usefulness. Second, verify that compared policies actually change the selected source or prompt. Third, when the claim concerns target-specific selection, compare against an assignment control that preserves the source distribution.

These checks prevent common inferential substitutions. Higher Hit@1 or correspondence cannot stand in for improved behaviour, and a retrieval-versus-no-memory contrast cannot isolate the value of the ranking rule. The appropriate baseline depends on the claim. No Memory tests access to the package, whereas a distribution-matched assignment control tests whether matching particular sources to particular targets adds value.

The P/C result adds a fourth baseline principle. When a retrieval-versus-empty comparison is interpreted as evidence for trajectory-specific content, a non-empty active control should preserve the generic interaction scaffold while removing the concrete trajectory. In this study, that control materially changed the conclusion: assignment remained practically small, but concrete coherent trajectory content had a large advantage over the abstract scaffold. Baselines should therefore be selected for the causal component named in the claim rather than for convenience alone.

## 6.6 Protocol integrity

The main study recorded 21 deviations involving service transport, runner recovery, replacements, and ledgers. Its 146 replacement executions followed an outcome-blind rule, six final episodes were missing, and no scientific result determined retention or rerunning. The first supplement retained 14 missing episodes and disclosed premature display of its condition map; no outcome was inspected and the frozen analysis code was unchanged. The P/C follow-up retained eight consumed-without-record executions, balanced four per arm, after two stale-run incidents. Its five-worker start stopped at a technical gate and resumed only after outcome-blind ledger amendments at four workers; all 800 targets remained evaluable, and the frozen materials and analysis hashes were unchanged. These events create procedural risk, but complete accounting, fixed analyses, scorer reconstruction, missingness sensitivity, and the absence of outcome-dependent rerunning reduce the likelihood that they explain the findings.

Auditability does not erase these deviations; it makes their possible influence inspectable. The important protection is that technical recovery rules were separated from scientific outcomes and that unresolved failures remained missing rather than being selectively replaced after unblinding.

## 6.7 Limitations and future work

The main study and first supplement lacked a protocol-only active control because N and N2 used empty wrappers. The later P/C follow-up directly addressed that omission for one prespecified control: C preserved a non-empty block, action–observation chronology, broad phase structure, approximate length, and a successful ending, while P supplied the concrete trajectory. P’s +20.06-point advantage shows that this abstract scaffold cannot fully reproduce the concrete trajectory package benefit.

This follow-up does not remove every content confound. C is deliberately more abstract and less semantically coherent or actionable than P, while P jointly adds concrete commands, observations, domain information, naturalness, and replay-verified success. P–C therefore estimates the deployed contrast between a concrete coherent trajectory and this particular scaffold, not a pure effect of experience or replay verification. Because the follow-up had no concurrent empty-memory arm, it also does not estimate C–N or permit arithmetic decomposition of R–N or P–N2 across collection windows.

A structural limitation is that candidate coverage was nearly saturated: 3,121 entries represented only 136 recipe signatures, and an exact signature-level match existed for essentially every target. This

availability made it feasible for D to saturate its internal criterion, but R’s 44-of-675 match count shows that it did not make ordinary raw selection trivial. The conclusion is bounded to a regime with abundant matching candidates; retrieval may matter differently when exact matches are rare or the procedure space is substantially larger.

External validity is limited by one hosted Qwen policy, one prompt, Top-1 retrieval, a frozen bank, and a simplified two-ingredient CookingWorld. Complete cookbook visibility may reduce the value of exact matching; structure-aware retrieval could matter more when the target procedure is latent or incomplete. The binary correspondence measure deliberately discards degrees of partial resemblance, and future work should test whether chain overlap, edit distance, or a separately validated graded similarity measure predicts realised utility more effectively. Conversely, a stronger policy might need less trajectory scaffolding. The findings may not transfer to agent-generated experience banks, whose successful trajectories can contain exploration errors, detours, and policy-specific noise absent from the environment-generated gold demonstrations used here.

Hosted-model stochasticity also limits fine-grained mechanism claims. Two repetitions provide a target-level mean but leave substantial uncertainty about individual targets, so the thesis does not label isolated positive or negative differences as transfer events. Replication across model snapshots and providers is needed before treating small target-level differences as stable.

The first supplement isolates assignment only within one frozen source vector and one derangement. It does not test uniform full-bank sampling, repeated independent permutations, or unsuccessful trajectories. The P/C follow-up inherits the same fixed P source vector and reuses all 800 targets. Consequently, the combined evidence supports a small assignment increment and a large concrete-trajectory increment relative to one abstract scaffold under this realised mapping; it does not show that arbitrary trajectories are interchangeable, that source success caused the effect, or that the findings replicate on an independent target sample.

As specified in Section 4.8.4, assignment intervals condition on one realised derangement and do not represent variability across admissible mappings; the result is not a generic estimate of random retrieval.

Target overlap means that agreement between R–N and R2–N2 is best viewed as numerical stability across collection windows, not replication on a fresh task sample. The same restriction applies to the numerical proximity of P–N2 and P–C. The 125 reserve targets add new coverage to the first supplement but do not dominate either follow-up estimate. Stronger evidence would use entirely new

targets, multiple independently generated permutations, and more than one active-control construction.

The primary estimands are average paired differences over the frozen evaluated target sets under one hosted-model configuration; extrapolation to new seeds, banks, or models is not licensed by the reported intervals.

The failure and overlap analyses were operationalised after outcomes were visible and remain descriptive. Future work should use fresh targets, repeat assignment controls, test other acting models and harder environments, and study domains where the relevant procedure is not already explicit in the semantic query. None of the present findings establishes that the agent performs—or fails to perform—human-like analogical reasoning.

Finally, the five-point margin encodes the practical effect chosen for this study, not a universal threshold. Smaller assignment benefits may matter at scale or in higher-value tasks. Conversely, equivalence within this margin should not be transferred to environments where a mismatched trajectory can cause irreversible real-world harm. The contribution is methodological and local: retrieval quality claims should match the intervention actually realised and the outcome that matters in deployment. Section 4.8.4 gives the study-specific decision rationale for this boundary.

## **7. Ethical and Broader Considerations**

### **7.1 Data, participants, and model-service exposure**

The study uses procedurally generated CookingWorld tasks and contains no human participants, personal data, user conversations, or behavioural profiles. The experience bank consists of synthetic cookbooks and environment trajectories. Prompts sent to the hosted policy service contain only these generated task materials and the fixed agent instruction; API credentials and operational secrets are excluded from the research records and release package.

The absence of human data limits privacy and direct participant risk, but it does not make the experiment consequence-free. The policy is accessed through a third-party hosted service, whose internal inference infrastructure and provider-side energy use are not observable. Model identifiers, request metadata, retry events, and returned token counts are therefore retained so that service exposure and resource use can be audited to the extent allowed by the provider interface.

## 7.2 Compute and environmental reporting

TextWorldExpress execution, trajectory replay, parsing, and BGE-M3 embedding ran locally; policy inference was provider-served. The original main collection produced 5,394 accepted records from 5,400 planned executions, of which 4,044 belong to the retained N/R/D analysis; the first supplement produced 4,786 of 4,800; and the P/C follow-up produced 3,192 of 3,200. Across the three formal collections, 13,372 of 13,400 planned executions produced accepted records. The frozen main-and-first-supplement accounting comprised 233,514,585 provider-recorded input tokens, 51,193,398 provider-recorded output tokens, 22,103 tokens from one overwritten call without a recoverable input–output split, and 61,911,200 conservative reserve tokens for interrupted calls without immutable usage records, totalling 346,641,286 tokens. Its USD 554.63 total comprised USD 455.57 supported by recorded API use or the overwritten call's recorded total and USD 99.06 in conservative reserve exposure. P/C accounted for an additional 103,515,869 conservatively accounted tokens and USD 165.63, including smoke and reserves for eight consumed calls without immutable usage records. The cumulative conservative accounting is therefore 450,157,155 tokens and USD 720.26. These totals combine recorded provider usage with explicit conservative reserves and should not be interpreted as invoice-level energy or carbon measures.

Token counts and monetary cost are imperfect environmental proxies. The provider does not disclose per-request energy use, hardware allocation, data-centre power effectiveness, or carbon intensity, so this thesis does not estimate emissions from unsupported assumptions. Resource use was constrained by prospectively fixed episode, token, and monetary ceilings. Top-1 prompting limited context expansion, and the post-result GR3–GR13 programme was cancelled rather than used to generate further paid evidence after the formal question had already been answered.

## 7.3 Broader implications

Experience memory can improve an agent without changing model weights, but a retrieved success may still be irrelevant or misleading in a new context. The present results add a complementary caution: an inspectable retrieval metric can improve dramatically without improving terminal behaviour. Systems deployed in higher-stakes domains should therefore evaluate retrieved context through downstream task outcomes and should verify that each claimed retrieval component actually changes the material shown to the acting model.

This experiment is not a safety validation for memory-augmented agents. CookingWorld contains no personal, medical, legal, financial, or physical-world consequences, and its official success signal is far clearer than objectives in most deployed systems. The finding that raw-semantic experience was

beneficial here does not justify automatically replaying stored trajectories in settings where obsolete, biased, private, or adversarial records could affect decisions. Experience provenance, deletion, access control, and human oversight remain outside the tested scope.

## 7.4 Code and reproducibility materials

The public reproduction repository is <https://github.com/whitesungun876/retrieval-alignment-realised-utility>. It contains the frozen protocols, environment and model manifests, task partitions, experience-bank hashes, retriever outputs, prompt and source bindings, P/C scaffold materials, deviation registers, outcome-only sanitised records, analysis outputs, analysis code, and figure-generation scripts. Provider raw response payloads, API credentials, signed service URLs, emails, transient worker state, and local service logs are excluded. The immutable hashes referenced in Appendix A identify the exact frozen materials used for the reported analyses.

## 8. Conclusion

This thesis asked why retrieved experience improved terminal success in TextWorldExpress CookingWorld. The complete Raw Semantic package produced a +20.59-point advantage over an empty block. De-lexicalisation changed almost every selected source and saturated its representation-internal criterion as designed, but its terminal-success increment over Raw Semantic retrieval was only +0.89 points (95% CI  $[-1.78, +3.63]$ ). The internal alignment manipulation therefore activated without accounting for the large package benefit.

Two sequential follow-ups, each frozen before its own outcomes, narrowed that benefit. A distribution-matched permutation retained most of the Raw Semantic package effect; the estimated target-specific assignment difference was +1.375 points (95% CI  $[-1.25, +4.06]$ ) and met the frozen  $\pm 5$ -point equivalence criterion. The later P/C follow-up found that a concrete coherent trajectory exceeded the paired abstract scaffold by +20.06 points (95% CI  $[+17.13, +23.06]$ ).

The strongest supported conclusion is therefore component-specific. In this high-redundancy bank, changing from mostly non-corresponding R selections to criterion-saturating D selections contributed little additional utility, whereas concrete coherent environment-generated gold-trajectory content contributed substantially relative to the implemented abstract scaffold. This does not show that any arbitrary trajectory would work, that replay-verified success caused the difference, or that the three studies form a single planned causal decomposition. Nor does it establish transfer to agent-generated experience banks containing exploration noise. It shows how sequential active controls can move an explanation from a retrieval-score story toward the content actually delivered to the agent.



The alignment–utility gap remains an entry point rather than the endpoint of the thesis. Its scientific value is that an internal criterion could be saturated while external behaviour remained nearly unchanged; the subsequent assignment and active-control studies then identified which explanations survived stronger controls. Generalisation is limited by complete cookbook visibility, one hosted model and prompt, a highly redundant bank, reused follow-up targets, one realised permutation, and one abstract scaffold construction.

The most informative next steps are replication with another acting model and fresh targets, comparison with multiple independent assignment controls and more naturalistic active scaffolds, direct manipulation of trajectory verification or outcome status, and structural retrieval where the relevant procedure cannot already be reconstructed from the semantic text. These extensions remain separate studies rather than post hoc changes to the completed protocols.

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# Appendix A. Protocol and Reproducibility Details

## A.1 Frozen retrieval implementation

The exact dense-retrieval configuration is given below. The same encoder artefacts and numerical settings are used for R and D.

| Parameter                 | Frozen value                                                                                         |
|---------------------------|------------------------------------------------------------------------------------------------------|
| Encoder                   | BAAI/bge-m3                                                                                          |
| Hugging Face revision     | 5617a9f61b028005a4858fdac845db406aefb181                                                             |
| Software                  | FlagEmbedding 1.4.0; transformers 4.57.6; tokenizers 0.22.2; torch 2.8.0; numpy 2.0.2; Python 3.9.25 |
| Representation            | Dense vector only; 1,024 dimensions; no sparse or ColBERT vectors                                    |
| Pooling and normalisation | CLS pooling; L2 normalisation                                                                        |
| Precision and device      | CPU float32; FP16 and BF16 disabled; deterministic PyTorch algorithms                                |
| Maximum input length      | 8,192 tokens; formal truncation prohibited (observed maximum: 65 tokens)                             |
| Prefixes                  | None for queries or sources; symmetric encoding                                                      |
| Batch size                | 8                                                                                                    |
| Similarity                | Cosine similarity implemented as inner product of L2-normalised vectors                              |
| Dense tie-break           | Label-blind SHA-256 rank after full-precision cosine score                                           |

**Table A.1. Frozen BGE-M3 dense-retrieval parameters.**

Ingredient names in D are replaced with role\_1, role\_2, and so forth according to first appearance in the cookbook; target–source correspondence labels are not used. Formal embeddings, rankings, and Top-1 selections are stored with content hashes.

## A.2 Deterministic source selection and ties

The initial source inventory used the 5,000 TextWorldExpress train-fold seeds 24,000–28,999. For each seed, the official environment generated its gold action sequence with generateGoldPath=True. Protocol v3.2 then grouped byte-identical cookbooks and retained the shortest stored action sequence; remaining equality was resolved by invalid-action count (zero for these gold paths) and then by SHA-256 of the canonical action JSON. Byte-identical canonical trajectories were subsequently removed by the same rule. Each retained action sequence was executed again from a fresh environment reset with generateGoldPath=False. A source entered the final 3,121-item bank only

when this replay reached official success and the visible cookbook matched the frozen source cookbook byte for byte. The agent-facing record contains the replayed initial observation and chronological action–observation sequence, not a newly generated LLM trajectory.

| Frozen scope               | Descriptive quantity           | Value                                                                                                                                                                                                                    |
|----------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source bank                | Recipe-signature frequency     | 3,121 entries; 136 signatures; median 16 entries/signature; maximum 97                                                                                                                                                   |
| Source bank                | Top-10 signature concentration | 818/3,121 entries (26.21%)                                                                                                                                                                                               |
| Main 675 targets           | Signature coverage by bank     | 93 unique target signatures; 675/675 targets covered; top-10 target-signature share 46.22%                                                                                                                               |
| Supplemental 800 targets   | Signature coverage by bank     | 94 unique target signatures; 800/800 targets covered; top-10 target-signature share 45.50%                                                                                                                               |
| Main R selections          | Selected-source reuse          | 474 unique sources; 137 selected at least twice; maximum reuse 7. Distribution: 1 use: 337 sources; 2 uses: 96 sources; 3 uses: 25 sources; 4 uses: 11 sources; 5 uses: 4 sources; 7 uses: 1 sources                     |
| Supplemental R2 selections | Selected-source reuse          | 515 unique sources; 181 selected at least twice; maximum reuse 8. Distribution: 1 use: 334 sources; 2 uses: 117 sources; 3 uses: 43 sources; 4 uses: 11 sources; 5 uses: 5 sources; 6 uses: 3 sources; 8 uses: 2 sources |

**Table A.2. Descriptive and archival summary of recipe-signature concentration, target-signature coverage, and selected-source reuse. Values are direct counts from the frozen source bank, formal retrieval selections, supplemental target materials, and R2 mapping; no inferential analysis is introduced.**

The 2,000 formal candidates were test-fold seeds 52,000–53,999. Before online collection, P7 sorted them by SHA-256 of the literal string `study3-v3.2-formal-target-order|<task_id>` and selected the first 675 required by the frozen power design. Retrieval scores, structural-match diagnostics, pilot outcomes, and agent outcomes were not inputs to this ordering. The 500 reserve targets were test-fold seeds 54,000–54,499 in the ascending-seed order created at P1. The supplement retained all 675 main targets and took the first 125 entries of this already frozen reserve list. A pre-outcome audit verified that those reserve identifiers had not appeared in earlier result-bearing JSON artefacts.

Retrieval ties are resolved by the preceding-stage semantic score and finally by a hash derived from a frozen salt and canonical candidate content. Neither source deduplication, target selection, nor tie-breaking can access condition labels, candidate position, structural-match labels, or agent outcomes. Candidate-order permutations and repeated index builds must reproduce the same selected source.

### A.3 Agent service and technical failures

Formal requests use the Alibaba Cloud Model Studio international endpoint, a maximum model output of 2,048 tokens, and a 120-second timeout. At most two retries are permitted for transport errors, responses that time out before returning a valid result, or provider-declared transient failures. An invalid action index, poor action choice, or unsuccessful episode is not retried. The log retains the

provider-returned model identifier, request identifier, timestamp, token use, latency, and finish reason for each call.

#### A.4 Blinding and run control

Operational manifests used masked arm labels until technical acceptance was complete. Condition order was counterbalanced within target and interleaved across batches so that no condition was confined to one collection period. Monitoring during collection was limited to completion state, service errors, missing records, token use, and cost.

#### A.5 Superseded protocol record

Protocol v3.1 used BAAI/bge-small-en-v1.5 and a Top-40 structural shortlist. It was stopped after 76 of 2,800 planned episodes when the embedding backbone and shortlist depth were revised. Scientific outcomes had not been inspected. The 76 records, their manifests, and their hashes are retained as an audit record but excluded from target selection, power analysis, error analysis, and all v3.2 results.

#### A.6 Protocol deviations

The original formal deviation register contains 21 entries covering provider transport outages, runner termination and heartbeat interactions, technical replacements, ledger or adapter recovery, and limited exposure of operational metadata during fault investigation. No episode was retained, deleted, or rerun because of its scientific outcome. The revised manuscript retains the N/R/D records and their original technical adjudication; removal of the unused arm is a reporting decision, not a rerun or reclassification of outcomes.

#### A.7 Original protocol and disposition of the non-activated H condition

Protocol v3.2 contained four formal conditions (N/R/D/H). Its original research question asked whether the complete Hybrid pipeline improved terminal success relative to Raw Semantic retrieval; H–R was the sole confirmatory contrast, and the 675-target power design was calculated for H–R. The original Holm family comprised R–N, D–R, and H–D. The revised manuscript is a post-unblinding restructuring of the reporting; no frozen data, estimates, or analysis code were altered, and the original confirmatory result is reported in full below.

| Original item | Frozen role                  | Frozen descriptive estimate / inference | Disposition in revised manuscript      |
|---------------|------------------------------|-----------------------------------------|----------------------------------------|
| N — No Memory | Four-condition reference arm | Target-weighted success 63.33%          | Retained as the empty-memory reference |

| Original item         | Frozen role                                                                | Frozen descriptive estimate / inference                                                       | Disposition in revised manuscript                                                     |
|-----------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| R — Raw Semantic      | Four-condition semantic baseline                                           | Target-weighted success 83.93%                                                                | Retained; R–N remains prespecified secondary                                          |
| D — De-lexicalised    | Intermediate representation condition                                      | Target-weighted success 84.81%                                                                | Retained; D–R remains prespecified secondary                                          |
| H — Hybrid Structural | Original complete-pipeline condition                                       | Target-weighted success 84.15%; same source and byte-identical prompt as D on 675/675 targets | Excluded as an independent graph intervention; retained as archival A/A evidence      |
| H–R                   | Sole original confirmatory contrast; power design targeted this comparison | +0.22 pp, 95% CI [–2.44, +2.96], sign-flip p = .914 (unadjusted primary p)                    | Reported in full as the original confirmatory package result; not a graph-only effect |
| R–N                   | Original Holm family                                                       | +20.59 pp, 95% CI [+16.89, +24.22], Holm p < .001                                             | Retained as prespecified secondary memory-package contrast                            |
| D–R                   | Original Holm family                                                       | +0.89 pp, 95% CI [–1.78, +3.63], Holm p = .883                                                | Retained as prespecified secondary representation contrast                            |
| H–D                   | Original Holm family                                                       | –0.67 pp, 95% CI [–2.15, +0.81], Holm p = .883                                                | Reclassified as an identical-prompt A/A diagnostic; not a graph-effect estimate       |

**Table A.7. Archival original-protocol disposition. All estimates and p-values are copied from the frozen P10 first-analysis outputs. H–R was the unadjusted sole confirmatory test; only R–N, D–R, and H–D formed the original Holm family. This table is archival and does not introduce a new inferential analysis.**

## A.8 Reproduction package

The public reproduction repository listed in Section 7.4 is <https://github.com/whitesungun876/retrieval-alignment-realised-utility>. It preserves the frozen manifests, content hashes, retriever outputs, outcome-only sanitised records, analysis outputs, analysis code, and figure-generation code needed to verify the reported tables and figures. Provider raw payloads, API credentials, signed service URLs, emails, transient worker state, and local service logs are excluded from the public release.

## A.9 Supplemental protocol and concurrency amendment

The supplemental freeze package contains the 800-target manifest, the split between 675 earlier formal targets and 125 frozen reserve targets, the complete R2 source vector, the derangement-constrained P assignment, renderer and prompt hashes, masked run order, three reverse-balanced sequence assignments, retry and missing-data rules, power outputs, analysis code, and episode, token, and cost ceilings. The source-assignment audit verifies that P and R2 use the same source multiset and reuse counts, that  $P_i \neq R2_i$  for every target, and that no donor target is also its own recipient. Operation overlap, structural labels, task difficulty, and agent outcomes are absent from the permutation input.

The original supplemental runner limit was four concurrent workers. A dated amendment made before inspection of any supplemental scientific outcome authorised a temporary increase to eight

workers while leaving the target manifest, condition order, model, prompts, sources, analysis, and budget ceilings fixed. After three provider HTTP 429 responses, the prespecified safety rule restored the four-worker limit. Those three executions form a subset of the 14 technical-missingness records reported below and were not rerun. The ledger retains their target, masked arm, repetition, request time, HTTP status, and retry disposition without replacing them on the basis of outcome.

SUP-R5 completed blind technical acceptance over planned cells, bindings, model and prompt hashes, source assignments, retry compliance, missingness, token accounting, and cost. Of 4,800 planned formal episodes, 4,786 produced accepted records and 14 remained technical missingness: six in P, four in R2, and four in N2. All 800 targets retained at least one valid execution in each condition. After acceptance, the masked arms were decoded and the frozen SUP-R6 script was run once. The first-analysis output and analysis bundle are archived under SHA-256 hashes `b3597d151cb84612bb564553e1a9f15908abcc67a5882246c5398292f038c852` and `49d8d1e90cd0f1928da81799aa55b4c1f7d8fc3041c4339055fed2ad1a45e86c`, respectively. The independently checked contrast export has SHA-256 `b5688b8a5de77dfa301012f0b982b449a1782d88fa0fac606c72e967fc2c6d10`.

One dated deviation records that the condition map was displayed during a schema-validation check before the analysis bundle was frozen. No outcome, success rate, score, or effect estimate was inspected, and the prospectively frozen analysis code retained its hash. The deviation does not change the reported computation but represents premature disclosure relative to the planned masking sequence. The supplemental failure taxonomy is reported as post-unblinding descriptive because the presently available archive does not establish that the complete executable rules and their precedence order were included in the prospective freeze package.

| Condition | Wrong operation | Wrong allocation | role | Non-required/<br>completed<br>ingredient | Premature<br>consumption | Navigation/step<br>limit | All failures |
|-----------|-----------------|------------------|------|------------------------------------------|--------------------------|--------------------------|--------------|
| N2        | 456             | 9                |      | 46                                       | 98                       | 7                        | 616          |
| P         | 263             | 6                |      | 0                                        | 6                        | 2                        | 277          |
| R2        | 226             | 6                |      | 8                                        | 14                       | 1                        | 255          |
| All       | 945             | 21               |      | 54                                       | 118                      | 10                       | 1,148        |

**Table A.9. Post-unblinding descriptive failure taxonomy for valid failed supplemental episodes.** The first three columns are subcategories of `operation_to_ingredient_error`; together they account for 1,020 failures (88.85%). The row-level file `sup_r6_failure_taxonomy.csv` in the reproduction package retains the first irreversible action and the observed and expected operations for every classified failure.

The post-unblinding worst-case missingness sensitivity assigned every missing R2 execution failure and every missing P execution success, then reversed those assignments. The resulting R2–P

estimates ranged from +1.125 to +1.750 percentage points; the corresponding 90% target-level  $t$  intervals were  $[-1.09, +3.34]$  and  $[-0.46, +3.96]$ , both within the frozen  $\pm 5$ -point equivalence bounds.

## A.10 Exploratory main-study diagnostics

The analyses in this section were operationalised after unblinding and are descriptive. They are reproduced here to preserve the complete diagnostic record without interrupting the principal results.

| Failure category                            | Count | Share of failures |
|---------------------------------------------|-------|-------------------|
| Ingredient–operation error                  | 804   | 87.87%            |
| Premature consumption of an ingredient      | 99    | 10.82%            |
| Navigation failure or step-limit exhaustion | 12    | 1.31%             |
| Reversed within-chain operation order       | 0     | 0%                |

**Table A.10a. Deterministic failure taxonomy over the 915 failed episodes in the retained N, R, and D conditions.**

| Condition | Valid episodes | Failures | Ingredient–operation | Premature consumption | Navigation / step limit |
|-----------|----------------|----------|----------------------|-----------------------|-------------------------|
| N         | 1,347          | 494      | 404 (81.8%)          | 82 (16.6%)            | 8 (1.6%)                |
| R         | 1,348          | 216      | 198 (91.7%)          | 14 (6.5%)             | 4 (1.9%)                |
| D         | 1,349          | 205      | 202 (98.5%)          | 3 (1.5%)              | 0                       |

**Table A.10b. Main-study failure composition by retained condition; percentages are shares of each condition's failures.**

Across the retained N/R/D conditions, ingredient–operation errors accounted for 804 of 915 failures. From N to R, ingredient–operation errors fell by 15.30 percentage points of episodes, premature consumption by 5.05 points, and navigation or step-limit failures by 0.30 points.

| Operation-overlap stratum         | Targets | N success | R success | R–N      |
|-----------------------------------|---------|-----------|-----------|----------|
| Target requires no transformation | 58      | 81.9%     | 97.4%     | +15.5 pp |
| Zero operation overlap            | 104     | 93.8%     | 98.6%     | +4.8 pp  |
| Partial overlap                   | 317     | 54.7%     | 79.8%     | +25.1 pp |
| Full coverage                     | 196     | 55.6%     | 78.8%     | +23.2 pp |

**Table A.10c. Post-hoc stratification of R–N by source–target operation overlap. Operation overlap was not assigned, the strata differed in baseline difficulty, and no multiplicity-adjusted inference was performed.**

## A.11 P/C protocol-control follow-up

The P/C protocol, paired experience materials, blinded manifest covering 3,200 formal executions (1,600 target–condition cells), sealed bindings, analysis code, power report, static QA, smoke



acceptance, engineering amendments, and formal technical-acceptance report are retained under `protocol_control_supplement/`. The frozen scientific hashes were: analysis code `7558648d...b4a5d`; paired experiences `dd8b178c...97fd`; formal manifest `dd63ef8a...ee2`; and sealed bindings `6e0fb442...b1b5`. The scientific artefact hashes were unchanged at outcome-blind technical closure.

Static QA verified 800 targets and 515 distinct P-source trajectories, each paired with one deterministic C scaffold and reused across the target assignments. It also verified exact P/C shell parity after body substitution, matched turn and phase counts, deterministic rendering, and zero detected leakage of concrete source or target content into C. The outcome-excluded eight-execution paid smoke passed model, prompt, binding, truncation, schema, scorer, leakage, and parity checks before formal authorisation.

Formal technical acceptance recorded 3,192 immutable records and eight balanced consumed-without-record executions, with all 800 targets retained. Total P/C accounting was 3,208 paid starts including smoke and conservatively reserved unresolved calls, 103,515,869 tokens, and USD 165.6254, below the frozen caps. Scientific unblinding and the deterministic analysis occurred only after this outcome-blind acceptance. The resulting target-level CSV and JSON report are stored under `protocol_control_supplement/analysis_formal_v1/results/`.

#### **A.11.1 Example paired P and C materials**

This archival example is copied verbatim from the frozen paired-material file. It uses target `cookingworld:test:52691` and the medium-length paired source `cookingworld:train:26399` (15 trajectory steps). P contains 916 whitespace-delimited words and 1232 local proxy tokens; C contains 890 words and 1124 proxy tokens. The C/P word-count ratio is 0.9716, and the frozen C/P proxy-token ratio is 0.9123. These are descriptive material properties, not outcome analyses.

#### ***P — complete concrete trajectory***

Interaction record  
Format: chronological action-observation sequence

Task description:  
Concrete task specification is shown below.  
Gather all following ingredients and follow the directions to prepare this tasty meal.

Ingredients:  
flour  
pork chop

Directions:  
grill the pork chop  
prepare meal

Initial state:  
You are in the kitchen. In one part of the room you see a stove. There is also an oven. You also see a fridge that is closed. In another part of the room you see a counter that has a knife, and a cookbook on it. In one part of the room you see a dining table, that has nothing on it. There is also a cutlery drawer that is closed. You also see a trash can that is closed. In another part of the room you see a dishwasher that is closed. In one part of the room you see a dining chair, that has nothing on it.

To the South you see the pantry. To the East you see the backyard.

Step 01 phase:

inspect

Step 01 action:

look around

Step 01 observation:

You are in the kitchen. In one part of the room you see a stove. There is also an oven. You also see a fridge that is closed. In another part of the room you see a counter that has a knife, and a cookbook on it. In one part of the room you see a dining table, that has nothing on it. There is also a cutlery drawer that is closed. You also see a trash can that is closed. In another part of the room you see a dishwasher that is closed. In one part of the room you see a dining chair, that has nothing on it.  
To the South you see the pantry. To the East you see the backyard.

Step 02 phase:

acquire

Step 02 action:

take cookbook

Step 02 observation:

You take the cookbook.

Step 03 phase:

inspect

Step 03 action:

read cookbook

Step 03 observation:

Gather all following ingredients and follow the directions to prepare this tasty meal.

Ingredients:

flour

pork chop

Directions:

grill the pork chop

prepare meal

Step 04 phase:

acquire

Step 04 action:

take knife

Step 04 observation:

You take the knife.

Step 05 phase:

acquire

Step 05 action:

open fridge

Step 05 observation:

You open the fridge. The fridge contains a raw pork chop.

Step 06 phase:

acquire

Step 06 action:

take pork chop

Step 06 observation:

You take the pork chop.

Step 07 phase:

acquire

Step 07 action:

close fridge

Step 07 observation:

You close the fridge.

Step 08 phase:

navigate

Step 08 action:

move south

Step 08 observation:

You are in the pantry. In one part of the room you see a folding chair, that has nothing on it. There is also a shelf that has some flour on it.  
To the North you see the kitchen.

Step 09 phase:

acquire

Step 09 action:

take flour

Step 09 observation:

You take the flour.

Step 10 phase:

navigate  
Step 10 action:  
move north  
Step 10 observation:  
You are in the kitchen. In one part of the room you see a stove. There is also an oven. You also see a fridge that is closed. In another part of the room you see a counter, that has nothing on it. In one part of the room you see a dining table, that has nothing on it. There is also a cutlery drawer that is closed. You also see a trash can that is closed. In another part of the room you see a dishwasher that is closed. In one part of the room you see a dining chair, that has nothing on it. To the South you see the pantry. To the East you see the backyard.

Step 11 phase:  
navigate  
Step 11 action:  
move east  
Step 11 observation:  
You are in the backyard. In one part of the room you see a barbeque. There is also a workbench, that has nothing on it. You also see a patio chair, that has nothing on it. In another part of the room you see a patio table, that has nothing on it. In one part of the room you see a clothes line, that has nothing on it. There is also a garden, that has nothing on it. To the West you see the kitchen.

Step 12 phase:  
transform  
Step 12 action:  
cook pork chop in barbeque  
Step 12 observation:  
You grill the pork chop with the barbeque.

Step 13 phase:  
navigate  
Step 13 action:  
move west  
Step 13 observation:  
You are in the kitchen. In one part of the room you see a stove. There is also an oven. You also see a fridge that is closed. In another part of the room you see a counter, that has nothing on it. In one part of the room you see a dining table, that has nothing on it. There is also a cutlery drawer that is closed. You also see a trash can that is closed. In another part of the room you see a dishwasher that is closed. In one part of the room you see a dining chair, that has nothing on it. To the South you see the pantry. To the East you see the backyard.

Step 14 phase:  
complete  
Step 14 action:  
prepare meal  
Step 14 observation:  
Adding the meal to your inventory.

Step 15 phase:  
stop  
Step 15 action:  
eat meal  
Step 15 observation:  
You eat the meal. It is delicious.

Interaction outcome: success

## ***C — complete paired abstract scaffold***

Interaction record  
Format: chronological action-observation sequence

Task description:  
An abstract task specification is shown below.  
The episode has several abstract requirements whose concrete names and values are intentionally omitted. The worked record demonstrates how an agent can organise a multi-stage interaction without supplying task-specific content.

Initial state:

Step 01 phase:  
inspect  
Step 01 action:  
inspect the currently available information and identify what the episode requires  
Step 01 observation:  
The available information is updated, making the next procedural requirement explicit. This step reduces uncertainty before a dependent choice is made. Its result is used to organise the remaining phase sequence.

Step 02 phase:  
 acquire  
 Step 02 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 02 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 03 phase:  
 inspect  
 Step 03 action:  
 inspect the currently available information and identify what the episode requires  
 Step 03 observation:  
 The available information is updated, making the next procedural requirement explicit. This step reduces uncertainty before a dependent choice is made. Its result is used to organise the remaining phase sequence.

Step 04 phase:  
 acquire  
 Step 04 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 04 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 05 phase:  
 acquire  
 Step 05 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 05 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 06 phase:  
 acquire  
 Step 06 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 06 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 07 phase:  
 acquire  
 Step 07 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 07 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 08 phase:  
 navigate  
 Step 08 action:  
 change the current context in order to reach the next relevant part of the episode  
 Step 08 observation:  
 The interaction enters the intended next context and exposes the next relevant opportunity. The transition changes context while preserving the outstanding task requirements. The agent continues only after the relevant abstract context becomes available.

Step 09 phase:  
 acquire  
 Step 09 action:  
 perform the next acquisition or access operation required by the episode plan  
 Step 09 observation:  
 The required abstract resource or access state is now available for subsequent steps. The operation establishes a prerequisite used by one or more later turns. The observation confirms access without revealing any concrete resource identity.

Step 10 phase:  
 navigate  
 Step 10 action:  
 change the current context in order to reach the next relevant part of the episode  
 Step 10 observation:  
 The interaction enters the intended next context and exposes the next relevant opportunity. The transition changes context while preserving the outstanding task requirements. The agent continues only after the relevant abstract context becomes available.

Step 11 phase:  
navigate  
Step 11 action:  
change the current context in order to reach the next relevant part of the episode  
Step 11 observation:  
The interaction enters the intended next context and exposes the next relevant opportunity. The transition changes context while preserving the outstanding task requirements. The agent continues only after the relevant abstract context becomes available.

Step 12 phase:  
transform  
Step 12 action:  
apply the required transformation to the appropriate abstract task item  
Step 12 observation:  
The abstract task item now reflects the requested intermediate transformation. The operation is applied only after its abstract prerequisites have been satisfied. The resulting state records progress toward the final task configuration.

Step 13 phase:  
navigate  
Step 13 action:  
change the current context in order to reach the next relevant part of the episode  
Step 13 observation:  
The interaction enters the intended next context and exposes the next relevant opportunity. The transition changes context while preserving the outstanding task requirements. The agent continues only after the relevant abstract context becomes available.

Step 14 phase:  
complete  
Step 14 action:  
combine the completed intermediate results and signal task completion  
Step 14 observation:  
The required intermediate results have been assembled and the task is ready to terminate. Completion occurs after the required intermediate states have been established. The record now moves from task construction to terminal handling.

Step 15 phase:  
stop  
Step 15 action:  
perform the terminal action only after the task has been completed  
Step 15 observation:  
The episode reaches a successful terminal state and no further action is required. The terminal operation is not taken prematurely. After success is observed, the interaction stops rather than adding unrelated actions.

Interaction outcome: success

## Appendix B. Declaration of Generative AI Use

This declaration is included for transparency and is intended to complement any programme-specific declaration required through KUNet or Digital Exam. During the planning, implementation, analysis, and writing of this thesis, the author used OpenAI Codex as an interactive generative-AI assistance tool. Codex was used to propose and revise experimental code, protocols, analysis scripts, tables, figure captions, and manuscript prose; to diagnose implementation issues; and to check language and internal consistency. Qwen3.7-plus was used as the acting model studied in the experiments, as described in the methodology, rather than as an uncredited author or outcome assessor.

The author made the study-design decisions, authorised all paid experimental runs, reviewed and edited the generated material, verified factual and bibliographic claims against the cited or archived sources, and interpreted the results. Terminal outcomes and scores were produced by the frozen environment and analysis code; generative AI was not used to judge whether an episode succeeded. No AI system is listed as an author or treated as a scientific source. The author accepts full responsibility for the accuracy, originality, integrity, and final wording of the submitted thesis.